

Plenary Lectures

PL-1. Opening lecture: Clinical impact of genome sequencing in lymphoid neoplasms

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Lymphoid neoplasms are a heterogeneous group of tumors with different clinical manifestations and evolution. The genetic and molecular mechanisms are not well understood. The systematic application of next generation sequencing technologies in these tumors have provided the first comprehensive view of somatic mutations revealing a remarkable molecular heterogeneity. Chronic lymphocytic leukemia (CLL) is the most frequent leukemia of adults in the Western countries. Genomic studies have revealed a remarkable molecular heterogeneity with only few genes mutated in up to 10-15% of the patients and a relatively large number of genes recurrently mutated at low frequency. The mutated genes tend to cluster in different pathways that include NOTCH1 signaling, RNA splicing, processing and transport machinery, innate inflammatory response, telomere protection, DNA damage response and cell cycle control, among others. Some of these mutated genes provide relevant prognostic information and constitute targets new therapeutic agents. These genetic alterations are differentially distributed in clinical and biological subsets of the disease indicating that they may drive at least in part their heterogeneous evolution. Genomic studies in mantle cell lymphoma, diffuse large B-cell lymphomas, splenic marginal zone lymphomas and Burkitt lymphoma have revealed a similar complex landscape of somatic mutations targeting pathways differentially represented in each tumor subtype. On the contrary, other tumors have a different pattern with a commonly mutated gene, such as BRAF and MYD88, mutated in virtually all cases of Hairy cell leukemia and Waldenström macroglobulinemia, respectively. In summary, global genomic studies of lymphoid neoplasms are reshaping our understanding of these diseases and provide new perspective for a more individualized diagnosis and new potential therapeutic targets.

PL-2. Clinical Use of Cancer Biomarkers in Epithelial Ovarian Cancer: Updated Guidelines from The European Group on Tumor Markers (EGTM)

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The most important biomarker for epithelial ovarian cancer is CA 125. When assayed on a single occasion, CA125 has a low sensitivity for early stage epithelial ovarian cancer as well as a low specificity, especially among premenopausal women. CA 125 is not recommended as a screening test in asymptomatic women without a hereditary risk for ovarian cancer. The Risk of Malignancy Index including CA 125, transvaginal ultrasound examination, and menopausal status is recommended in differential diagnosis to estimate the probability of malignant potential of a pelvic mass. CA 125 is recommended as an alternative prognosticator when it is difficult to estimate the amount of multiple intra-abdominal metastases by imaging. CA 125 is also recommended for monitoring

activity of disease during treatment and follow-up. To obtain a CA 125 response, an above cut-off level concentration (>35 kU/L) should decrease $\geq 50\%$ (four measurements) or $\geq 75\%$ (three measurements). For patients with elevated pretreatment concentrations that normalize on first-line chemotherapy (<35 kU/L) or that never normalize, the suggested criterion for progression is doubling of remission nadir levels. Initial studies suggest an increased specificity of HE4 for ovarian cancer as compared to CA 125, mainly in premenopausal women. The sensitivity of HE4 and CA 125 for ovarian cancer however, appears to be similar. Given conflicting results in multiple trials, HE4 either alone or when used in the Risk of Ovarian Malignancy Algorithm is not recommended for routine clinical use.

PL-3. Guidelines for the differential diagnostic evaluations of effusions

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Effusions are a frequent reason for clinical consultations and about 5 - 20% of all cases are of neoplastic origin. Between 24 and 50% of patients with disseminated cancer develop pleural effusions, with about 100,000 new cases per year in the USA. These data clearly show the importance of having an efficient method for diagnosis (primary-metastatic) and differential diagnosis of patients with effusions. Cytology of the fluid is the main technique used, but unfortunately over 30% of malignant effusions display negative cytology.

This study presents a guideline for the diagnostic possibilities of malignant effusions, using tumor markers. This guideline describes the main problems in determination and interpretation of tumour markers in fluid effusions. In general, tumor markers may be classified into two groups; those normally released by mesothelium such as Mesothelin, CA 125 or CYFRA 21-1 and those tumor markers not released by mesothelium such as CEA, CA 15.3, CA 72.4 etc. Detection of high concentrations of tumor markers included in this last group indicated the presence of metastases. However, important discrepancies have been reported in relation to the cut-off, for example from 3 ng/ml to 50 ng/ml with CEA. To avoid this problem, the simultaneous determinations of tumor markers in effusions and serum have been suggested. If concentrations are higher in the effusion compared to serum it indicates local production and indirectly metastases. Using these criteria tumor marker sensitivity is high 70-80% depending on the chosen combination with a high specificity ($>98\%$) in both patients with negative or positive cytology in the fluid.

Another important clinical problem is the diagnosis of Mesotheliomas. Mesomark is the tumor marker suggested for this malignancy with significantly higher concentrations in patients with mesothelioma. Mesomark sensitivity is around 50-60% in mesothelioma, with the highest concentrations in epithelioid mesotheliomas. The specificities vary between 80 and 95%, with increases mainly in patients with lung and ovarian cancer, but also in some individuals with benign disorders, mainly those related to asbestos. Classical tumor markers are not very useful. However, abnormal CEA exclude mesothelioma where it is possible to find high concentrations of CYFRA 21-1 and less frequently CA 15.3.

PL-4. Biomarkers for cutaneous and uveal melanoma-Metastasis prediction

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Background: Cutaneous Malignant Melanoma patients have a poor survival in spite of new drug developments. Uveal Melanoma (UM) sends metastasis mainly to the liver. Therefore, it is essential to detect early liver metastasis formation and improve survival. Biomarkers assessing patients' response and prognosis are very important.

Aims: Elevated serum Biomarker levels in UM were previously demonstrated by us. We evaluated now a panel of Biomarkers: Osteopontin (OPN), S-100, Melanoma-inhibitory activity (MIA), Tissue Polypeptide-Specific Antigen (TPS), TK, VEGF, in Metastatic Cutaneous and UM vs Disease-Free (DF) patients (pts) aiming to study which Biomarker will best predict and earlier detect Liver Metastasis, thus enabling effective treatment with impact on pts. survival.

Methods and Patients: Levels of serum biomarkers were analyzed for 128 DF and 84 metastatic UM, as well as for 64 Cutaneous Malignant Melanoma pts. compared to 100 healthy Controls. A matched-pairs analysis was used to compare pre- and post-metastasis patients and treatment marker levels. Differences in biomarker levels were analyzed for each treatment group and correlated to survival and Metastasis free status.

Results: UM 69 pts underwent brachytherapy and 15 were enucleated. A significant decrease in S-100 levels in the enucleated pts (0.09 to 0.06 µg/l, $p = 0.0492$) after 1 m and in the brachytherapy pts (0.11 to 0.08 µg/l, $p = 0.0251$) after 10 m, were demonstrated. Significant increasing levels (5–25 fold) ($p = 0.002$) in the BioMarkers OPN, MIA, VEGF, TK and S-100 were predictive for liver metastases, as shown later by CT. In the Cutaneous Melanoma pts. increases in all markers indicated treatment unresponsiveness and recurrence, while significant decreases were correlated to PR to the new biological therapies.

Conclusions: Treatments of UM resulted in significant decreases in S-100, OPN, TK and VEGF levels. Significant increases in all markers detected earlier than CT and liver enzymes development of UM liver metastases. We introduced S100 as a routine biomarker for Cutaneous and UM pts follow up, to verify response to new biological drugs and reduce the number of CTs.

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PL-5. How to Validate a New Cancer Biomarker: From Discovery to Clinical Application

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Despite an enormous amount of research in recent years, few new tumor biomarkers have entered clinical use. This applies especially for serum biomarkers, with no such widely used biomarker having been introduced in the last 25 years. The reason for this decline in the availability of new biomarkers appears to be due to the increased demand from healthcare

payers and regulators, that new biomarkers deliver patient benefit in a cost-effective manner. In order therefore, to take a newly discovered putative biomarker to the clinic, it must undergo a series of rigorous evaluations. These validations included analytical validation (assessment of imprecision, analytical sensitivity, analytical specificity, linearity and potential interference), clinical validation (i.e., confirm that the results of the biomarker assay separates patient cohorts into different categories that will require different treatments or have different outcome), clinical utility (i.e., show that measurement of the marker results in a clinical decision that positively impacts on patient outcome relative to the current standard of care). Ideally, clinical utility should be established using a level I evidence study, such as a prospective clinical trial, retrospective analysis of archival specimens that were collected prospectively from subjects participating in at least 2 studies or from a meta/pooled analysis of retrospective/prospective trials. Finally, before entering the clinic, there is an increasing requirement that a new biomarker be shown to be cost effective and undergo regulatory approval.

PL-6. Moving forward to synthetic lethal approaches for improving survival in NSCLC

Rosell R

Non-small-cell lung cancer (NSCLC) is commonly diagnosed at the metastatic stage, with median survival of one year. The identification of driver mutations in the epidermal growth factor receptor (EGFR) in a subset of lung adenocarcinomas as the primary oncogenic event led to a model of targeted therapies and to the genetic profiling of NSCLC. Currently, the use of EGFR tyrosine kinase inhibitors (TKIs) confers remission in 60% of patients, but responses are still short-lived. The pre-existing EGFR T790M mutation could be a subclonal driver responsible for these transient responses. In addition, AXL overexpression and reduced MED12 function are hallmarks of resistance to TKIs in EGFR-mutant NSCLCs. Crosstalk between signalling pathways is another mechanism of resistance; therefore, the identification of the molecular components involved could lead to the development of combination therapies co-targeting these components in lieu of EGFR TKI monotherapy. In addition, novel biomarkers could be identified through deep sequencing analysis of serial rebiopsies before and during treatment.

Non-small-cell lung cancer (NSCLC) is the most frequent type of lung cancer and the most common cause of cancer death.¹ Poor survival in NSCLC is in part due to the development of drug resistance. Despite extensive research on resistance mechanisms, preclinical data has not been incorporated into the selection of patients and tailored treatment regimens in clinical trials. Because of this failure, we still lack an accurate picture of how efficient treatment can be in the right setting. Preselection studies are complicated by the fact that multiple different resistance mechanisms may occur, potentially influencing the optimal biomarkers to use depending on the specific driver gene mutation, a mutation that confers a selective growth advantage to the cell in which it occurs.² Until recently, few studies had attempted to preselect patients using specific biomarkers that could predict response to a specific treatment regimen.³ For example, in NSCLC patients with mutations in the epidermal growth factor receptor (EGFR), the response rate to erlotinib was 58%. Although remissions were transient, with progression-free survival (PFS) of 9–7 months⁴ this was a significant improvement over results with chemotherapy. Based on this data, the US Food and Drug Administration (FDA) has recently approved the use of erlotinib as first-line treatment in NSCLC driven by EGFR mutations.

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PL-7 Characterization of NSCLC cell lines for discovery of synthetic lethal therapies

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Despite the recent paradigm shift in the treatment of NSCLC patients, with a move toward biomarker-directed therapy, preclinical data has seldom been incorporated into clinical practice, since the existence of multiple resistance mechanisms complicates the selection of optimal biomarkers. Reversible or irreversible EGFR TKIs have offered a therapeutic alternative for patients with metastatic EGFR positive lung cancer but primary or acquired resistance limits their therapeutic success. Well characterized cell lines, may imitate gene expression profile of lung cancer patients and can be used as tools to test and identify rational synthetic lethality approaches. In 2004, Sordella and colleagues were able to demonstrate the differential EGF-induced tyrosine phosphorylation pattern seen with wild-type and mutant EGFR cell lines. Y845 which activates the JAK2/STAT3, pathway is highly phosphorylated in the L858R missense mutant cells, but not in the wild-type or the deletion mutant, and hence appears to be unique in distinguishing between the two types of EGFR mutations and sensitivity to EGFR TKIs alone or in combination with JAK2/STAT3 inhibitors, like ruxolitinib. SHP2 on the other hand has a fundamental role for NSCLC cells harboring EGFR mutations affecting the levels of the proapoptotic protein BIM. SHP2 Y542 phosphorylation is induced in the EGFR wild type H1666 cells in response to EGF but not in the H3255 cells, which harbor the missense L858R exon 21 mutation, suggesting that SHP2 activity may be less efficiently promoted by EGFR L858R. The mutant-bearing (del19) PTEN-null cell line H1650 does in contrast exhibit inducible SHP2 Y542 phosphorylation. The reduced SHP2 phosphorylation and full ERK activation may partially correlate with sensitivity to EGFR TKIs plus JAK2/STAT3 inhibitors, while the increased SHP2 activity may correlate with sensitivity to EGFR TKIs with MET, AXL, EphA2 or even SHP2 inhibitors. Following this line of research at the levels of cell lines or xenografts models we can understand better EGFR mutant NSCLC and contribute to the radical transformation of its treatment.

PL-8. Genetics and biomarkers in personalisation of lung cancer treatment

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First- line EGFR TKIs monotherapy for patients with mutant EGFR NSCLC is incorrect and EGFR inhibitors, reversible or irreversible, are

unlikely to provide cures in the majority of patients. The expression levels of the proapoptotic protein BIM have been found to predict responsiveness to kinase inhibitors in treatment naïve cancer patients, confirming that this molecule is implicated in the modulation of the cancer cell dependence on EGFR and other oncogene models. The levels of all three major splicing isoforms, BIM-EL, BIM long and BIM short, are induced after erlotinib treatment in drug-sensitive PC-9 cells but not in drug-resistant H1650 and in H1975 cells. However pretreatment BIM expression levels may not be enough to predict the outcome of EGFR TKIs. The two primary signaling pathways activated by EGFR include the MAPK, and the PI3K, axes. Src tyrosine kinases and the activation of the STAT3 pathway and downstream signaling have also been well documented. EGFR phosphorylation leads to the recruitment of multiple effector proteins through recognition and binding of SHP2 to phosphotyrosine motifs on the receptor. SHP2 is a ubiquitously expressed SH2 domain-containing protein tyrosine phosphatase. Despite its direct function in protein dephosphorylation, SHP2 plays an overall positive role in transducing signals initiated from growth factors/ cytokines and extracellular matrix proteins and initiating various downstream signaling cascades, including the PI3K, MAPK and STAT3. We can simply define three groups of patients: 1) Patients with low BIM levels at baseline who can be treated with the combination of EGFR TKIs with compounds that downregulate or abrogate the activity of SHP2. 2) Patients with high BIM levels at baseline, that may have hyperactive the JAK/STAT pathway through the L858R mutation, or through the loss of SHP2 activity, or even through the presence of the EGFRvA isoform. This group of patients can be treated with the combination of EGFR TKIs plus a JAK2 inhibitor; 3) Patients with high mTOR levels, which can be translated to a higher dependence of the tumour on the Warburg effect or the glutamine pathway and can be targeted with the combination of EGFR TKIs with drugs that interrupt glycolysis or glutaminolysis.

PL-9. New rules, new drugs: Structural insights in developing compounds against the EGFR T790M gatekeeper resistance

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The better knowledge of molecular mechanisms underlying NSCLC has made possible to identify tyrosine kinases (TKs) as promising therapeutic targets involved in a variety of cancers, due to their role in the modulation of growth factor signaling. Historically, many therapeutic strategies were used: 1) blocking the kinase activity of these receptors by designing novel Tyrosine Kinase Inhibitors (TKIs) or 2) inhibiting the corresponding cytoplasmic second messengers involved in cell signaling pathways. The use of target therapies listed above permitted the development of several approved drugs (e.g. gefitinib or erlotinib).

Unfortunately, prognosis of NSCLC depends nowadays not only on histological features but also on genetic mutation profiles. Several molecular alterations have been identified as driver mutations in NSCLC, which are responsible for acquired resistance to TKIs. Particularly, activating mutations of Epidermal Growth Factor Receptor (EGFR) are mainly represented by deletion in exon 19, substitution of methionine by threonine at amino acid position 790 of EGFR gene domain in exon 20 (T790M) and the L858R substitution in exon 21. Among these drug resistance causing factors, the secondary point mutation T790M becomes one of the main factors involved in acquiring resistance to EGFR reversible inhibitors in NSCLC targeted treatments.

The application of molecular modeling techniques in designing new EGFR inhibitors was encouraged by the elucidation of the molecular

structure of EGFR experimentally. This information promoted the application of structure-based drug design-techniques (SBDD) for the design of novel drug-resistant inhibitors and to study their mechanism of action, prompting substantial efforts to discover a second generation of EGFR inhibitors that can overcome the effects of the T790M substitution.

This presentation will point out the different strategies followed by medicinal chemists to overcome T790M drug resistance and how computer-aided drug design strategies may help to design novel and more efficient inhibitors.

PL-10 Generation of orthotopic mouse models of lung cancer suitable to investigate new therapeutic strategies

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Lung cancer (LC) is a devastating disease and a major therapeutic burden with poor survival rates. It is responsible of around the 30% of all cancer deaths. Lung cancer is classified in two mainly groups: small-cell lung carcinoma (SCLC) (20%) and non-small-cell lung carcinoma (NSCLC) (80%). Patients with SCLC are mainly treated with chemotherapy, and only a small minority of patients achieves long-term survival. In NSCLC the identification of several genetic alterations, as presence of EGFR mutations (15%) and translocation EML-ALK (2-7%) has permit the identification of novel effective and therapies for several patient subgroups. Thus, therapeutically improvement must come for the personalization of patient treatment by categorization in function of their genetic profile. Today, although DNA sequence information from tumors has revealed a number of frequently occurring mutations, many of the relevant driver mutations remain undefined. Therefore, the establishment of preclinical models phenocopying patients' primary tumor features, which accurately reflect phenotypic, genotypic, and tumor chemotherapy response behavior, is a basic step to identify novel therapeutic targets and treatments. Here we showed the development of several tumor models by the orthotopically engraftment of fragments of histologically-intact primary lung tumors (named *orthoxenografts*®) in nude mice. Eight of 12 (67%) primary LC tumor were implanted/perpetuated in mice. Extensive local tumor growth and regional and distant metastases were observed. Approach also useful for the perpetuation of lung tumors derived from genetically-engineered mouse models (GEMMs). Thus, we showed evidences that *orthoxenografts*® of lung cancer may be outstandingly valuable preclinical models for new drug development, reducing the high failure rate that exists in translating preclinical results to patients, but mainly for the development of personalize oncology strategies.

PL-11. Light-regulated stapled peptides to inhibit protein-protein interactions involved in clathrin-mediated endocytosis

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Most biological processes are mediated by protein-protein interactions (PPIs), which have emerged as important pharmacological targets [1].

Peptide inhibitors with nanomolar affinities were recently developed and demonstrated anticancer activity in vivo [2]. In order to dissect the precise orchestration of PPIs in cells, it would be very useful to complement the pharmacologic selectivity of inhibitors with a means of remotely controlling their kinetics and site of action. Here, we present a peptide design and screening method to obtain photo-switchable inhibitors of protein-protein interactions, and we apply it to photo-regulated clathrin-mediated endocytosis (CME) in living cells [3]. The selective manipulation of CME with light using these peptides, named Traffic Lights (TL) because they act as stop & go signals for membrane traffic, constitutes a novel tool to control cell signaling in spatiotemporally defined patterns. TL peptides can be applied to dissect the role of CME in complex cellular functions like receptor internalization, cell growth, division and differentiation.

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PL-12. Retrospective Analysis on the Diagnostic Capacity of Biomarkers in Oncology

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The few nowadays known organ specific biomarkers like prostate specific antigen (PSA), human calcitonin (HCT) or the monoclonal immunoglobulin are able to support diagnosis of prostate cancer, medullary carcinoma of the thyroid or multiple myeloma. In addition those tumor markers are helpful to establish diagnosis whose increased release coincides frequently with a special type of cancer: alpha-fetoprotein (AFP), human chorionic gonadotropin (HCG), progastrin-releasing peptide (ProGRP), HER-2/neu shed antigen and S100. AFP is the biomarker of first choice for hepatocellular carcinoma – if a germ cell tumor in younger men with retroperitoneal lymph node metastases can be excluded. A release of ProGRP > 200 pg/ml is a strong indicator for small cell lung cancer. A release of HER-2/neu shed antigen > 50 ng/ml is leading the way to primary breast cancer, although his release into blood must not be correlated to a HER2-overexpression in the metastases. Likewise the release of S100 protein > 1 ng/ml is a strong indication for metastatic malignant melanoma – if a significant bacterial infection or a neurodestructive or neurodegenerative process can be excluded. Beside these single tumor markers possessing high diagnostic capacity the degree of release of further markers is helpful in diagnostics. Here the leading marker is relevant (like cancer antigen (CA)125 and HE4 in ovarian cancer, CA 19-9 in pancreatic cancer, SCCA in squamous cell carcinomas) and also the non release (like CA15-3 in colorectal cancer).

Hence in the situation of CUP and in contrary to all other indications in oncology the determination of all tumor markers mentioned here in parallel is important (CEA, AFP, CA125, human epididymis protein 4 (HE4), CA19-9, CA72-4, CA15-3, cytokeratin fragment (CYFRA)21-1, neuron-specific enolase (NSE), ProGRP, SCCA, S100, HER-2/neu shed antigen, HCT, and free light chains and immunoglobulins in bone metastases as well as PSA in men).

PL-13. Computer assisted classification of hormone receptor expression in breast cancer in relation to cell shape

van Rodijnen N and Nap M

Hormone receptor status in breast cancer is an important support factor for therapeutic decisions. Objective assessment of the level of expression is difficult. Computer assisted quantification of nuclear staining (Histoquant, 3Dhistec) on digital whole slide images (WSI) provides a partial solution. There is still the need to identify the area of infiltrative tumor interactively in each WSI.

We have used WSI of H&E stained tissue sections to identify areas of infiltrative tumor. A tissue micro array (TMA) robot was used to select tissue cores from these areas and combine them in a TMA. 4 tissue cores from up to 10 tumors were collected in one tissue block and serial sections were cut and incubated according to standard protocols with antibodies against ER and PR. Counterstaining was done with hematoxylin. Stained slides were then digitised and now staining was analysed in every core without interactive selection. In parallel every core was given a manual score from 0–4 and 6, as was the usual procedure.

The Histoquant data were compared with the manual scores and fed to a principal component analysis (PCA) procedure. It turned out that for every record 3 parameters were essential to classify every core in the best possible way. These parameters were blue color intensity in the nucleus, brown color intensity and the nuclear perimeter and surface, joined in a shape factor.

Records with a shape factor equal or above 1 were considered single cells and included in the calculations. Projection of these three parameters in a 3-dimensional space resulted in clustering of the data points that showed a strong relation with the traditional manual scores. Introduction of the shape factor of the nucleus improved the separation of different classes. These results show that the addition of morphological findings to computer assisted scoring of hormone receptor expression in breast cancer refines the classification in an objective procedure with good correlation to conventional, but subjective methods.

PL-14. Autoantibodies as an aid to cancer detection?

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Early detection and diagnosis of many cancers is paramount to improve prognosis by enabling early treatment prior to the onset of metastatic disease. The literature widely documents the presence of an immune response, in the form of IgG antibodies raised to tumour-associated antigens (TAAs) in cancer patient's serum. Autoantibodies may provide an *in vivo* amplification of carcinogenesis, months to years before the tumour bulk becomes otherwise clinically detectable, and their measurement may be used as an aid to early diagnosis in a number of solid tumours.

We have analysed the autoimmune response in the sera of patients with (in particular) lung and liver cancer at the time of diagnosis, and compared the results to those obtained from age and gender matched normal sera and sera from individuals at a higher risk of developing the disease. We have also looked for these antibodies in a few individuals where sera were available before the cancer had presented clinically.

Results illustrate the value of panel-antigen vs single-antigen mediated autoantibody detection, and suggest that an optimised TAA panel for different cancers would improve on current gold-standards serving as a cost-effective means to improve prognosis of cancers worldwide.

The detection of antibodies in sera of some individuals prior to clinical symptoms of disease, in both lung and liver cancer, suggests that this technology may also be useful as a means of risk stratification, and therefore closer surveillance of such individuals.

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Middleton et al. Serum Autoantibody Measurement for the Detection of Hepatocellular Carcinoma: Opportunities for Early Tumor Identification? Submitted

PL-15. Immunogenic cell death Biomarkers in the prognosis and outcome prediction of cancer patients

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Background: Cell death products released as a consequence of cancer disease or cytotoxic therapy may lead to sustained stimulation of immune cells in the tumor microenvironment. Serum levels of these immunogenic biomarkers may bear great potential to assess patient prognosis and therapy outcome.

Patients and methods: In prospective trials on a total of 221 cancer patients undergoing local therapies (49 patients with secondary liver metastases from colorectal cancer treated by SIRT, 50 patients with primary liver cancer treated by TACE) and systemic cytotoxic chemotherapies (44 patients with breast cancer and 78 with pancreatic cancer), levels of proimmunogenic HMGB1 and nucleosomes and antiimmunogenic soluble RAGE were measured in serum samples taken before and during treatment. They were compared with radiological therapy responses and patient survival.

Results: In cancer patients, HMGB1 and nucleosome levels were higher than in control groups, and RAGE levels were decreased. Further, HMGB1 and nucleosome levels rapidly increased 24 hours after application of local SIRT or TACE therapies, whereas RAGE levels decreased further. In CRC patients undergoing SIRT, elevated HMGB1 and nucleosome values were associated with poor therapy response and poor overall survival. In liver cancer patients undergoing TACE, high nucleosome and low RAGE levels were associated with insufficient therapy response and unfavourable prognosis. In breast cancer patients, pretherapeutic high HMGB1 and nucleosome levels and low RAGE values indicated poor response to neoadjuvant chemotherapy. In pancreatic cancer patients undergoing primary chemotherapy, high nucleosome and HMGB1 levels and low RAGE levels during therapy were associated with poor survival.

Conclusion: Proimmunogenic cell death serum marker HMGB1, nucleosomes and antiimmunogenic soluble RAGE are valuable biomarkers for the prediction of therapy response and prognosis in cancer patients.

PL-16. The Roche HE4 Assay - Results from the HE4 Multicenter Evaluation

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BACKGROUND: HE4 is a new, valuable marker for the diagnosis and management of ovarian cancer.

OBJECTIVE: To evaluate a new HE4 assay adapted to the Elecsys® system.

MATERIAL AND METHODS: The analytical and clinical performance of the HE4 (Roche Diagnostics) assay was evaluated at three different European sites as compared with the Elisa system (EIA) from Fujirebio Diagnostics. We evaluated the sera of 537 healthy and 110 pregnant women, 484 patients with benign gynaecological disorders and 264 patients with ovarian cancer.

RESULTS: Total imprecision (%CV) of Elecsys HE4 ranged between 2.3 and 8.7%; inter-laboratory reproducibility between 3.4 - 5.1%. Linearity upon dilution showed a mean recovery of 98.5% (SD +9.1%). The overall correlation for the 1,395 samples was $r = 0.9783$ with a slope of 1.8%. The median of HE4 serum levels in healthy individuals was 54.9 pmol/L, the 95th percentile 96.5 pmol/L. The concentrations were continuously increasing with age and race, with significantly lower concentrations ($p < 0.01$) in black Americans (47 ± 13 pmol/L) than in Caucasian (56 ± 20 pmol/L). Pregnant women had significantly lower concentrations as compared to healthy women with similar age. Using the recommended cut off value of 120 pmol/L HE4 levels were higher than this value in 4% of patients with benign diseases and in 81% of patients with ovarian cancer. By contrast CA 125 was higher than the 95th percentile of healthy individuals in 18% and 83% of these patients, respectively. Concerning ovarian cancer HE4 was clearly related to tumor stage and histological subtype with significantly higher concentrations in patients with serous (398 ± 96 pmol/L) and the lowest levels in mucinous tumors (101 ± 27 pmol/L) ($p < 0.001$).

CONCLUSIONS: Elecsys HE4 provides accurate methodological characteristics and demonstrates a good clinical correlation with HE4 EIA from Fujirebio. Our data show the need for age and race dependent reference values for single investigations.

PL-17. HE4 and its value for the gynecologic oncologist - first results from the Berlin-ROMA study

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This multicentric prospective study aimed to determine the role of HE4 and CA125 in combination with ultrasound parameters in predicting the presence of ovarian cancer in pelvic mass patients. All patients received a systematic ultrasound examination according to IOTA criteria. At the

same point blood and urine samples were collected. Ultrasound was performed by gynecologists or experts in ultrasound.

The primary end-point of the study is to develop a new diagnostic algorithm for ovarian cancer based on HE4, Ca125 and ultrasound parameters. Secondary aim points are to analyze if HE4 and Ca125 are stable in urine samples, and if their values correlate with serum concentrations; as well as the value of HE4 in therapy monitoring in patients with ovarian cancer.

We planned to enrol 870 patients, of whom 218 patients will be expected to present with a malignant ovarian tumor.

Patient enrollment started in August 2013. 250 patients have so far been enrolled in the study, representing the discovery cohort. Data on these 250 patients will be presented

PL-18. HE-4 utility to increase efficiency in patients with abdominal masses

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BACKGROUND: Ovarian cancer has vague symptoms that explain why most patients are diagnosed at an advanced stage. CA 125 is the tumor marker of choice in this malignancy but has low sensitivity and specificity, mainly in premenopausal women. Recently another tumor marker, HE4 has been described with similar sensitivity and higher specificity. Likewise, the algorithm, ROMA (Risk of Ovarian Malignancy Algorithm) is suggested to improve the sensitivity and specificity of CA 125 & HE4 by combining both tumor markers in patients with abdominal masses.

METHODS: CA125, CA 19.9 and HE4 serum levels were determined in 989 consecutive women who arrive to our hospital with abdominal masses. Final diagnosis was non-malignant in 799 patients (233 endometriosis, 325 ovarian cysts and 241 with other diseases), 45 patients with ovarian tumors with low malignancy and 145 patients with ovarian cancer. CA 125, CA 19.9 and HE4 cut-offs were 35 U/mL, 37 U/mL and 120 pmol/L, respectively. ROMA algorithm cut-off was 13.1 and 27.7 for premenopausal or postmenopausal women, respectively.

Results: HE4, CA 125 and ROMA results were abnormal in 2%, 24% and 27% of patients with benign diseases, and in 85, 84% and 93% of patients with ovarian cancer. Both tumor markers, HE4 and CA 125 were related to tumor stage and histological type, with the lowest concentrations in mucinous tumors. By contrast CA 19.9 was the most sensitive marker in mucinous tumors. A significantly higher area under the curve was obtained with ROMA and HE4 as compared to CA 125 in the differential diagnosis of benign gynaecological diseases versus malignant ovarian cancer (0.947, 0.942 and 0.849 respectively).

CONCLUSIONS: Data from our population indicate that ROMA algorithm do not increase efficacy in premenopausal women and only slightly in postmenopausal women, with a similar results to using HE4 and CA 125 alone. These results are still more evident when patients with mucinous tumors were excluded.

PL-19. HE4 is an excellent biomarker for the management of women with endometrial cancer

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Introduction: Endometrial cancer is the most common gynecological malignancy with over 280,000 cases worldwide each year. For women

diagnosed with endometrial cancer biomarkers are critical for preoperative assessment, detection of progressive disease and prognosis.

Discussion: HE4 is over expressed in patients with endometrial cancer. Moore et al showed that median HE4 serum levels are significantly elevated in all endometrial cancer stages relative to the healthy subjects. When considering stage I patients alone, median HE4 levels are significantly elevated compared with controls. As well, a multi-marker analysis revealed that HE4 has greater sensitivity for all stages of disease compared with CA125.

HE4 is a valuable biomarker for the pre-operative assessment of patient with endometrial cancer. Moore et al examined the correlation of HE4 levels to myometrial invasion in patient with endometrial cancer. The median serum HE4 level for stage I patients with no myometrial invasion was 45.0 pM, significantly less than the median serum level of 100.0 pM ($p = 0.002$) for stage I patients with myometrial invasion. In addition, patients with stage IA disease had significantly lower median serum levels than patients with more advanced stages. Examination of the performance characteristics of HE4 for predicting stage IA versus stage IB disease revealed a sensitivity of 94.1%, a specificity of 56.1% and a NPV of 97.0%.

Moore et al assessed the accuracy of serial measurements of HE4 in patients undergoing chemotherapy and monitoring for recurrence. Using a cut-point of 25% for the velocity of change the accuracy of HE4 for predicting progression was 81.4% with a NPV 91.7%. For patients in the monitoring phase the mean lead time for detection of progression was 39 days. HE4 is also a prognostic marker. Brennen et al, showed that HE4 is a indicator for recurrent disease and survival. A Kaplan–Meier analysis showed HE4 levels were associated with impaired recurrence free survival (RFS) and overall survival (OS) for patients with elevated HE4 levels.

Summary: The novel biomarker HE4 is expressed in patients with endometrial cancer and is a powerful marker for assessing preoperative myometrial invasion, response to chemotherapy, detecting progressive disease and for prognosis.

PL-20. Human cancer epigenetics: from knowledge to application

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For the last twenty-five years an increasing amount of evidence has shown the relevance of epigenetics in cell biology and tissue physiology, being DNA methylation aberrations in cancer the flag-ship for the recognition of its disturbance in human diseases. From the candidate gene approaches, new powerful technologies such as comprehensive DNA methylation microarrays and whole genome bisulfite sequencing has recently emerged that have reinforced the notion of epigenetic disruption in the crossroad of many sickness. From the poster-boy cases of *MGMT* and *GSTP1* hypermethylation in the prediction of alkylating drug response and prostate cancer detection, respectively, to the personalized treatment of leukemia with small molecules targeted to fusion proteins involving histone modifiers such as DOT1L and MLL, the field has walked a long path. The current talk will focus in the epigenetic profiling, basically at the level of DNA methylation and histone modifications, that is starting to provide clinical value in the diagnosis, prognosis and prediction of response to drug therapies, with an emphasis in neoplasia, but without forgetting the novel advances in other human disorders. For cancer, we have already a wide view of the undergoing DNA methylation events that expand beyond classical promoter CpG islands of tumor suppressor genes and we have a growing list of mutated chromatin remodeler genes that contributes to the tumorigenesis process. It is time to apply this knowledge in practical clinical situations like the diagnosis

of cancers of unknown primary, the screening of malignancies in high-risk populations or a biomarker selection of the patients that should receive treatment with epigenetic drugs.

PL-21. Epigenetic deregulation in glioma, biomarkers and new opportunities

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The presentation will review the predictive value of *MGMT* promoter methylation in glioblastoma, the most malignant form of primary brain tumors (glioma, WHO grade IV) for benefit from alkylating agent therapy. This is a mechanistically plausible relationship due to the DNA repair function of *MGMT* that removes the most cytotoxic lesion introduced by alkylating agents. Surprisingly, a strong prognostic effect was observed for *MGMT* methylation in clinical trials with grade III glioma. Analyses of the genetic and epigenetic context of this surprising, and clinically relevant difference revealed major differences. In glioblastoma we found *MGMT* methylation usually associated with deletion of Chromosome 10, on which *MGMT* resides (10q26). In contrast, glioma grade III are characterized by gain of function mutations in the isocitrate dehydrogenase 1 (*IDH1*) gene. The mutant form results in a neomorphic function producing the oncometabolite 2 hydroxyglutarate, which inhibits epigenetic modifier proteins such as TET2. This perturbation yields a CpG island methylator phenotype (CIMP, >70%) that includes *MGMT* promoter methylation. Furthermore, in glioma grade III CHR10 deletions are infrequent, in particular in CIMP+ cases. Expression analyses suggest that grade-III gliomas do not completely shut down *MGMT* expression. The prognostic effect in glioma grade-III may rather be linked to other CIMP deregulated genes that await discovery.

PL-22. Development of epigenetic biomarkers for urological cancer detection and prognosis assessment

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The most common genitourinary (GU) neoplasms (i.e., prostate, bladder, and kidney cancers), are generally clinically silent at their earliest stages, when curative treatment is most likely successful. However, there are no consensual guidelines for GU cancer screening and available methods are characterized by suboptimal sensitivity and specificity. Moreover, standard clinical and pathological parameters meet with important limitations in the assessment of prognosis on an individual basis. Epigenetically based cancer biomarkers are promising tools for early detection and diagnosis, assessment of prognosis, and prediction of response to therapy. The importance of early diagnosis in Oncology has been emphasized and although current methodologies (such as cytology, histopathology, immunohistochemistry, etc) play a critical role, molecular markers may complement or, eventually, replace them if more efficient. Because epigenetic alterations often precede the emergence of the malignant phenotype, they are advantageous for early cancer detection. They might also be detected in clinical samples, including those obtained by exfoliative cytology, endoscopic brush techniques, and biopsy, as well as detection

in urine, saliva, stool, plasma and sputum samples, allowing for non- or minimally invasive molecular detection of tumors. Moreover, early detection of cancer relapse before it manifests clinically (or on imaging during routine patient follow-ups) might be accomplished by analyses of epigenetic-based biomarkers. Some epigenetic alterations have also been associated with tumor behavior and may thus serve as biomarkers for prognosis. Among the cancer-related epigenetic alterations so far reported, aberrant DNA methylation is, by far, the most widely studied, followed by deregulated expression of chromatin machinery proteins and microRNAs.

In my presentation I will focus on the recent studies performed by our research team regarding the development of epigenetic-based GU cancer biomarkers.

PL-23. A pre-clinical breast cancer stem cell model based on MMTV-PyMT mice for testing nanoparticle directed therapy

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The stem cell model of cancer predicts that cancer treatment failure can be overcome if treatment is directed to the self-renewing cancer stem cell (CSC). Most therapeutic approaches are directed to the proliferating cells of the tumor and not the tumor stem cells. In order to advance therapy to tumor stem cells, we need models of cancer in which the stem cells can be identified, targeted and treated. Cells comprising the microenvironment (stem cell niche) of CSCs also play a defining role in optimal tumor initiation potential and are a second target for treatment. We used the MMTV-PyMT transgenic mouse model of spontaneous breast cancer to identify both CSCs and tumor supporting mesenchymal cells. Female mouse mammary cancer (FFMC) cell lines consisting of two major populations were generated from primary breast cancers: CD24⁺CD29⁺CD44^{low}CD49f⁺Sca-1^{low} TICs and CD24⁺ tumor-derived mesenchymal stem cells (TDMSCs). CD24⁺ 419II cells at 1×10^3 consistently initiated mammary gland tumors, whereas the CD24⁺ TDMSCs failed to initiate such tumors at 1×10^7 cells. Co-transplantation of TDMSCs and CD24⁺ TICs resulted in a significant increase in TI to the level of a single CSC. Knock-down of Bmi-1, a polycomb group protein and stem cell transcriptional promoter, by shRNA prevented 419II cells from initiating tumors. TI was restored when Bmi-1 was reintroduced via an expression vector. We now propose to use anti-CD49f coated nanoparticles containing shRNA to inhibit Bmi-1 and/or docetaxol to target therapy to the CSCs. Preliminary experiments are now underway to determine localization of anti-CD49f coated particles to growing breast cancers transplanted to syngeneic FVB/N mice. Future experiments are planned to direct therapy to the TDMSCs as well as the CSCs.

PL-24. Molecular profiling of bladder cancer: from –omic signatures to individual candidates

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Bladder cancer is a molecular disease, resulting from the consecutive accumulation of genetic, epigenetic and environmental factors, among others. We are investigating the molecular alterations involved in

tumorigenesis and progression at the genetic, epigenetic and protein levels covering several independent aspects of these alterations using in vitro, in vivo, or clinical material. The integration of these analyses will accelerate the interpretation of the biological significance of such critical events and facilitate a molecular sub-classification of the different pathologic and clinical subgroups. In this scenario, our group has been working over the past years on characterizing whether hypermethylation of profiles, focused panels of tumor suppressor genes (TSGs), and individual candidates could adjunct an early detection and classify bladder cancer subgroups according not only to pathologic or clinical criteria but also to patient outcome. Part of our studies is aiming at evaluate the clinical utility of CpG hypermethylation for the diagnosis and molecular classification of bladder tumors based on their aggressiveness and clinical behaviour: from comprehensive CpG array methylation profiling, to several multiplexed panels of TSGs, and finally characterizing specific individual methylated candidates. In this talk, the major results of these three strategies for bladder cancer diagnosis and sub-classification will be summarized: using CpG arrays, methylation specific multiplex ligation-dependent probe amplification (MS-MLPA), and methylation specific polymerase chain reactions (MS-PCR) analyses. Validation and functional analyses will finally reveal the biological role of the new candidates towards novel biomarker utilities or as potential therapeutic targets.

PL-25. Pharmaceutical significance of alkaline phosphatase as a drug target and a protein therapeutic

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Advancements in our understanding of the biological functions of alkaline phosphatase (AP) isozymes in recent years has led to efforts to translate that knowledge into novel therapies either using recombinant APs as protein therapeutics or targeting AP function itself via small molecule modulators. In this lecture I will summarize both therapeutic approaches related to bone AP. Bone AP, also known as tissue-nonspecific AP (TNAP) encoded by the human ALPL gene, is essential for bone mineralization. The deposition of bone mineral, hydroxyapatite, normally only takes place in the fibrillar collagenous matrix of the skeleton and in teeth while calcification of the extracellular matrix of soft tissues is prevented by the omnipresence of the potent mineralization inhibitor inorganic pyrophosphate (PP_i). The expression of high levels of TNAP in mineralizing tissues leads to the hydrolysis of PP_i, and thus a change in the P_i/PP_i ratio to values that are permissive for normal controlled mineralization. Hypomorphic mutations in the ALPL gene lead to the heritable genetic disease hypophosphatasia (HPP) where accumulation of PP_i and reduction in the P_i/PP_i ratio causes rickets/osteomalacia. Increased expression of TNAP in bone as a result of osteoblastic metastases to bone or Paget's bone disease leads to an increase in the P_i/PP_i ratio and excessive, inappropriate deposition of bone mineral in the skeleton, known as bone sclerosis, which can be extremely painful. Ectopic over expression of TNAP in vascular smooth muscle cells leads to sinking levels of PP_i in the vasculature and pathological ossification of the arteries. This is prevalent in the rare disease generalized arterial calcification of infancy (GACI), but also common in patients with chronic kidney disease, type II diabetes, and obesity. We have developed mineral-targeting TNAP as enzyme replacement therapy that, by normalizing PP_i levels, prevents all the manifestation of life-threatening HPP. We have also developed pharmacological inhibitors of TNAP that inhibit the inappropriately elevated levels of TNAP in the vasculature in mouse models of GACI that have the potential to be effective in models of osteoblastic metastases and Paget's disease.

PL-26. Receptor for advanced glycation end products in patients with kidney cancer

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Background: The receptor for advanced glycation end-products (RAGE) and its ligands – advanced glycation end-products (AGEs), S100 proteins and high mobility box group protein 1 (HMGB1) are involved in the pathogenesis of cancer. Glyoxalase I (GLO I) is an enzyme which detoxifies AGE precursors.

Study Aims: The aim of the study was to find out the relationship between four polymorphisms (SNP) of the RAGE gene (AGER) and one SNP of GLO I gene and clear cell renal cancer (ccRCC).

Methods and Materials / Patients: All polymorphisms (RAGE -429 T/C, -374 T/A, 2184A/G, Gly82Ser and GLO A419C) were determined by PCR-RFLP in 214 patients with ccRCC. A group of 154 healthy subjects was used for comparison.

Results: We have found significant differences in the allelic and genotype frequencies of GLO I A419C SNP between patients and controls - higher frequency of the mutated C allele in ccRCC - 58.6% vs. 44.5% in controls, OR (95% CI) 1.77 (1.32-2.38), $p < 0.001$, OR (95% CI) CC vs. AA 2.76 (1.5-4.80), $p < 0.001$, AC + CC vs. AA 2.03 (1.23-3.30), $p = 0.005$. High aggressiveness of the tumour (grade 4) was associated with the presence of mutated alleles RAGE -429 T/C and RAGE 2184A/G SNPs ($p = 0.001$ and $p < 0.001$ mutated vs. wild type alleles, respectively for genotypes $p = 0.008$ a $p = 0.005$, respectively).

Conclusions: Our results demonstrate the link of A419C GLO I SNP to the presence of the tumour and the connection of RAGE -429 T/C and 2184 A/G SNPs with the aggressiveness of the tumour. Further studies are required, especially with respect to potential therapeutic implications.

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PL-27. HE-4 in the diagnosis of abdominal masses. Comparison with ultrasonography and RMI

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Efficiency in the preoperative assessment of adnexal masses is essential. An incorrect assessment of the risk of malignancy may lead to inappropriate medical and surgical procedures. Surgical strategies for treatment and staging of ovarian cancer require specific skills. Additionally, prognosis will depend on the type of surgery performed. Nowadays, there are different complementary tests for the assessment of pelvic masses: ultrasound and variants (doppler, 3-D, logistic regression models), NMR, TC, tumor markers (CA124, HE4, panels of markers) and models of variables integration, as RMI (risk of malignancy index). Until now, none of the tests used to assess the risk of malignancy has been enough sensitive. Actually, based on the findings, many of the non-malignant tumours should be considered "at risk" preoperatively. This circumstance implies mismanagement of patients, mobilizing oncology resources that are expensive and non cost-efficient. Recent advances in ultrasound, tumor markers and multivariate risk index designs as "ROMA" among others,

seek to change this scenario. In this lecture the contribution of ultrasonography and tumours markers in the differential diagnosis of pelvic masses are discussed, emphasizing the recent proposals.

PL-28. HE-4 levels in gynaecological patients undergoing surgical treatment for suspected malignancies. Systems to increase efficiency

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BACKGROUND. Ovarian cancer is a malignant disease with high mortality and morbidity. Diagnostic systems are not enough efficient and frequently surgery is performed without a real knowledge about the diagnosis. Patients with cancer need the collaboration of different specialists, including the pathologist, during large and complicated surgery. In contrast some patients with benign diseases are treated in a similar way, when laparoscopy could be sufficient.

OBJECTIVE. To evaluate tumor marker utility presurgically, and to establish risk of malignancy

METHODS: CA125 and HE4 serum levels were determined in 631 consecutive women who arrived to our hospital with abdominal masses. Final diagnosis was non-malignant in 511 patients (191 endometriosis, 281 ovarian cysts and 239 with other diseases), 32 patients with ovarian tumors of low malignancy and 88 patients with ovarian cancer. CA 125 and HE4 cut-offs were 35 U/mL and 120 pmol/L, respectively. ROMA algorithm cut-off was 13.1 and 27.7 for premenopausal and postmenopausal women, respectively.

RESULTS: HE4, CA125 and ROMA results were abnormal in 2%, 21% and 25.8% of patients with benign diseases, and in 82, 82% and 91% of patients with ovarian cancer. Both tumor markers, HE4 and CA 125 were related to tumor stage and histological type, with the lowest concentrations in mucinous tumors. Abnormal HE-4 indicated a high probability of malignancy: 98% in women older than 50 years, 88% in women between 40-50 years and 80% in women younger than 40 years with non-mucinous tumors. In contrast positive predictive value for ROMA was 91%, 49% and 35% in these three groups, respectively. It is interesting to point out that patients with HE-4 concentrations lower than 70 pmol/l had a probability of cancer lower than 2%.

CONCLUSIONS. Data from our patient group indicate that tumor markers may be useful tools to evaluate cancer risk and determine type of initial surgery.

PL-29. Making Use of the ROMA Algorithm - A Gynaecological Oncologist's Perspective

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The adoption of novel biomarkers, such as HE4, into routine clinical practice often depends on the familiarity, availability and first-hand experience of these new technologies within the region, area or country. This presentation will focus on the applications of CA125/HE4 and Risk of Ovarian Malignancy Algorithm (ROMA) through the perspective of real-life scenarios. Specifically, with particular focus on the interpretation, medical decision and patient management of several clinical cases, while assessing the importance of biomarkers CA125 and HE4 in clinical decision making in patients presenting with a pelvic mass.

We will discuss key considerations, including surgical approach, medical equipment that must intervene and level of the institution that should perform the procedure, all of which may be directly impacted by integrating an oncologic biomarker evaluation within the context of the clinical evaluation. These tools ultimately, enable the physician/surgeon gain additional objectivity in decision-making.

PL-30. HE-4 in ovarian cancer therapy monitoring: comparison with CA 125

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Background. The main clinical utilities of tumor markers are follow-up and therapy monitoring. CA 125 is the tumor marker of choice in ovarian cancer, but another new tumor marker, HE-4 has been reported recently. The advantages to use several tumor markers in the follow-up have been only scarcely studied.

Objective: to compare the clinical utility of CA 125 and HE-4 in therapy monitoring of patients with ovarian cancer.

Methods: Serial CA125 and HE4 serum levels were determined in 119 patients treated for ovarian cancer during a period of two years. All patients had a minimum of 4 determinations during a period of at least 6 months. Therapy responses were calculated using the criterias of the EORTC. Treatment was chemotherapy, using different chemotherapy regimens. CA 125 and HE4 cut-offs were 35 U/mL and 120 pmol/L, respectively. We considered as response when pretherapy tumor marker serum levels decrease to at least 50% in two serial determinations during at least 3 weeks. Similar criterias were used to indicate progression: increase of at least 50% in two serial determinations.

Results: CA 125 was correlated with tumor response in 83%, HE-4 in 85% and any of them in 92% of our patients. Similar results were found when mucinous tumors were excluded: any of the tumor markers was useful in 96% of our patients. It is interesting to indicate that CA 125 was better than HE-4 (higher concentrations, previous changes, etc.) in 33% of our patients, HE-4 better than CA 125 in 29% of patients and similar in the remaining 30% of patients with ovarian cancer. However, it is interesting to notice that changes in the predominant tumor marker appeared in 25% of patients during therapy.

Conclusions: Both tumor markers may be used in the follow-up of patients with ovarian cancer during therapy. The reason to explain changes in the tumor marker pattern of the tumors could be further evaluated

PL-31. Liver cancer: Framing the major issues for clinical decision making

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The diagnosis and treatment of patients with liver cancer has significantly improved in the last two decades. Joint efforts by experts from different fields and geographical origin have elaborated clinical practice guidelines to be used for the management of these patients. Thus, current clinical management of HCC patients is shared by practicing physicians and investigators. This allows further research to build on top of this clinical ground.

The current challenges affect from molecular profiling to diagnosis, prognosis prediction and treatment. Novel “omics” technologies will ultimately lead to a molecular classification that is linked to treatment decision. However, biopsy sampling has to face the heterogeneous nature of the tumor within the nodule, across nodules and also, along its evolution. Hence, major effort should be placed in the fruits of the so-called “liquid biopsy” from peripheral blood. This may also facilitate an earlier diagnosis or the recognition of the disease even prior to overt malignant transformation. Until then, imaging techniques will be instrumental to improve the current detection rate as well as the diagnostic confirmation capacity. Improved performance and applicability of screening ultrasound are needed to reach an optimal effectiveness of surveillance, but imaging techniques have to gain further accuracy for staging. Molecular imaging may become a reality.

The improved knowledge in all aspects should change the current selection of the first line treatment approach that is mostly based in rough assessment of tumor size and number. Criteria for liver transplantation and surgery may be modified if it would be feasible to accurately predict the risk of recurrence and the potential to reduce/eliminate it by molecular targeted agents. Locoregional approaches based in physical damage may be slowly replaced by biology-based treatments acting selectively against tumor cells or the surrounding stromal population. Sorafenib has been the first agent to show that tumor progression may be halted with a significant survival increase, but there is need for major improvement to finally transform cancer into a chronic condition.

Combination of laboratory and clinical research with generous and critical insight from all sides is the sole way to advance in all these fronts, while not forgetting that the optimal approach would be to be able to prevent HCC development.

PL-32. Biomarkers for HCC

Biomarkers may be used for surveillance prior to the diagnosis of hepatocellular carcinoma, for staging or characterization, or for monitoring progress after therapy. Some biomarkers may be useful in all three situations. The first biomarker for HCC was alpha-fetoprotein (AFP). Initially, in the era prior to the availability of ultrasound and CT scanning it was a useful diagnostic test. More recently it has found its widest use as a surveillance tool. However, the evidence favouring its use in this situation is weak, with the test showing relatively poor sensitivity although high specificity. Thus despite widespread use it is not recommended by two of the three major professional liver disease societies (AASLD and EASL). Other biochemical tests have been used, including the L3 fraction of AFP, which is more sensitive than AFP, and des-carboxygamma prothrombin (DCP), also known as PIVKA II. These two markers, (and AFP) are markers of advanced or aggressive disease. They therefore cannot also be markers of early disease, and therefore should not be used for surveillance. Although all three markers are frequently identified as being elevated in patients who have established HCC this does not mean that they are useful for surveillance. DCP has better sensitivity than AFP, but neither DCP nor AFP-L3 have been adequately tested as surveillance markers. All three markers have been used to monitor patients after therapy. It has been documented that sometimes elevation of these markers precede the appearance of radiologically detected HCC recurrence. However, it has never been shown that the use of these markers has any effect on outcomes, i.e., that the use of these markers improves survival. Tissue biomarkers may be useful in selecting treatment. To date there is only one example. High levels of c-met in tissue may identify patients who respond well to oral met inhibitors, such as tivantinib. No other tissue biomarker has yet been shown to be useful.

PL-33. Molecular characterization of hepatocellular carcinoma

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Liver cancer is the second cause of cancer mortality and a major health problem globally. Only one molecular therapy, sorafenib, has been approved for advanced cases. There is limited understanding of the pathogenesis of the disease. The field-effect predisposing to HCC development is characterized by activation of signaling pathways related to IL-signaling, oxidative stress, EGF signalling and inflammation among other. Mutations in promoter regions of TERT have been identified in 25% of cases in preneoplastic lesions. Also inactivation of p53 in stellate cells appears to be a potential gatekeeper. During the last decade and after applying next generation sequencing several drivers have been identified. Each HCC contains around 35–40 mutations, among which 6–8 are considered drivers. The main mutations described are in the promoter region of TERT, p53, CTNNB1, ARID1A and Axin 1. Afterwards the landscape of mutations is characterized by a long tail of mutated genes in <5% of cases, such as RAS, PI3K and others. In terms of high level amplifications at 5–10% prevalence containing oncogenes are in 11q13 Cyclin D1 and FGF19 and 6p21 VEGFA, whereas other amplifications described contain Myc and Met among others. All this molecular information should be directed to select populations for proof-of concept trials, such as the one testing MEK inhibitors in RAS + patients, or trial enrichment, such as the one testing MET inhibitors in MET –positive populations.

PL-34. Clinical use of tumor markers in China

The workshop will highlight the current status of cancer in China and the important role of oncology biomarkers in screening, risk stratification, therapy selection, disease progression, and monitoring for relapse.

Cancer is the leading cause of death in China, and cancer mortality is projected to increase significantly over the next 20–30 years. Tan Weiyun has reported that every minute six Chinese are diagnosed with cancer on the mainland and every minute five people die from the disease [1]. A survey conducted in 2012 by the Shanghai Municipal Center for Disease Control and Prevention (SCDC) showed that on an average 1.79 persons among 100 in the city are cancer patients. In 2011, the average was 1 person in 100. The SCDC survey reveals that every day, 82 people die of cancers in Shanghai. According to the survey, the top 10 fatal cancers for men are lung, large intestine, stomach, liver, prostate, pancreas, esophagus, urinary bladder, kidney, brain and central nervous system. The leading cancers in women are breast, large intestine, lung, stomach, thyroid gland, liver, pancreas, brain and central nervous system, gall bladder and ovary. These tumors account for more than 85 percent of all the cancers. Several researchers have investigated the cause of cancers. Wang has reported that chronic infection is the main risk factor for cancer in China, accounting for 29.4% of cancer deaths (31.7% in men and 25.3% in women), followed by tobacco smoking (22.6% with 32.7% in men and 5.0% in women), low fruit intake (13.0%), alcohol drinking (4.4%), low vegetable intake (3.6%) and occupational exposures (2.7%) [2]. Laboratorians in China play a vital and leading role in triaging, diagnosing, and managing patients who visit the hospitals with symptoms related to various cancers. They rely on oncology biomarkers to assist the physicians and patients by providing useful information by means of least invasive procedures such as a simple blood test.

The experts will present role several key markers such as Pepsinogen I/II which are used for screening and diagnosis of atrophic gastritis which can

lead to gastric cancer. The topics of discussions will also include risk stratification and triage for patients suspected of lung cancer with Lung Cancer Biomarker Panel comprising of SCC, ProGRP, CEA, and CYFRA 21-1; ovarian cancer with ROMA comprising of HE4 and CA 125.

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PL-35 Lung cancer screening

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Lung cancer screening has been one of the most controversial areas in Medicine for many years. Studies done in the 1970s failed to show any benefit from screening with chest radiography in terms of lung cancer mortality. Thereafter, screening has been recommended against by all professional associations and medical institutions worldwide for decades. In 1999, a study called Early Lung Cancer Action Program was published. It showed that early detection of lung cancer was possible using low dose computed tomography (LDCT) of the chest. This study resulted in a new interest in research on lung cancer screening and was followed by a larger international multicenter study called I-ELCAP. Results showed that 85% of lung cancers can be diagnosed in Stage I resulting in a 10 year survival rate in excess of 80%. A large randomized controlled study in the US (NLST) was halted ahead of time in 2010 because it had already shown that screening with LDCT significantly reduces mortality. Since then many favourable recommendations by different specialty associations have been published in the last couple of years. However, in spite of the changing tide the debate regarding lung cancer screening continues and is now centered around risk profiles, cost-effectiveness, the need for biomarkers, potential harms, and smoking cessation.

PL-36. The Roche ProGRP assay for differential diagnosis in lung cancer - results from the MCE study

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BACKGROUND: The Elecsys® progastrin-releasing peptide (ProGRP) immunoassay was designed to support differential diagnosis in lung cancer using serum and plasma samples. We conducted a multicenter evaluation of the assay across Europe and China. **METHODS:** Technical and clinical performance of the Elecsys® ProGRP assay was evaluated at three European sites and two Chinese sites by means of imprecision, stability, method comparison, and differentiation potential. **RESULTS:** Intermediate imprecision across five analyte concentrations ranged from 2.2% to 6.0% coefficient of variation. Comparable recovery was observed with plasma and serum samples at various storage conditions. There was good correlation between the Elecsys® and ARCHITECT assays in plasma samples (slope 1.00, intercept 0.02). The Elecsys® ProGRP assay also showed good correlation between serum and plasma samples (slope 0.95, intercept 2.75). ProGRP differentiated small cell and non-small cell lung cancer in serum (AUC 0.90, 95% CI: 0.87 to 0.93; 78% sensitivity, 95% specificity), with no clinically relevant effects of site, age, gender, or smoking habits. ProGRP also showed good differential diagnostic capacity in serum from patients with estimated glomerular filtration rate (eGFR) ≥30 at all sites. ProGRP concentrations were low in patients with benign diseases (median 40 pg/mL), or other malignancies (median 40 pg/mL). **CONCLUSIONS:** Increased stability of the Elecsys® ProGRP

assay in serum and plasma offers clear benefits over existing assays. This first evaluation of a ProGRP assay in Chinese patients demonstrated reproducibility of results among different ethnicities, with no clinically relevant differences in ProGRP levels between European and Chinese patients.

PL-37. The unmet medical needs in lung cancer and what laboratory medicine can contribute to diagnosis and therapy management of lung cancer

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Lung Cancer is one of the most common malignancies in males and females, and although it is the fourth most common cause of cancer, it has the second highest death rate. It accounts for 20% of all cancers in men with a life time risk of 1 in 13 and 12% of all cancers in women with a life time risk of 1 in 23. Worldwide lung cancer incidence is estimated at 1.6 millions with about 1,095,200 cases in males and 513,600 in females. The global burden of lung cancer has increased dramatically and reflects the smoking epidemic in the developing world.

The initial investigations are usually a chest radiograph, a CT scan of the thorax and upper abdomen. This usually determines if there is a suspicious pulmonary opacity that requires further investigation. Diagnosis usually requires sampling of the abnormal area. It is important to consider diagnosis and staging together and the abnormal area, which would confer that higher staging, should be biopsied whenever possible.

The use of laboratory markers like CEA, CYFRA or NSE as a prognostic and predictive marker in patients with lung cancer is widely debated. Most of the recent published guidelines do not recommend the use of such markers. On the other hand, there is a clear need for better prediction of the prognosis and the response to specific therapy. Therefore future studies may benefit from evaluating more than one marker at a time, which may possibly create a more precise index for prognosis and recurrence of lung cancer, that would be possible by the use of single biomarkers.

PL-38. Tumor Markers in lung cancer; a tool for early diagnosis?

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BACKGROUND: Lung cancer (LC) is the most frequent and fatal human cancer. Its prognosis is directly related to early diagnosis. In this study, we hypothesized that a combined panel of six serum tumor markers (TM) is useful in the diagnosis of patients with clinical suspicion of LC and evaluated the possible utility in early diagnosis.

METHODS: We investigated the diagnostic utility of a combined panel of six serum TM (CEA, CA125, SCC, CYFRA 21-1, NSE and ProGRP) in 3,644 consecutive individuals referred to our institution because of the clinical suspicion of LC.

RESULTS: Using standard clinical procedures, LC was confirmed in 2,259 patients (1,876 with NSCLC (83%) and 383 with SCLC (17%)) and excluded in 1,429 individuals (38%). The sensitivity of the TM

panel investigated here for the diagnosis of LC was 90.0%, its specificity 81%, its NPV 85% and its PPV 89%. In patients with early stages of LC these figures were, respectively, 73%, 81%, 91% and 69% for NSCLC stage I/II and 96%, 81%, 99% and 36% for intra-thoracic SCLC. Finally, two TM (NSE and ProGRP) significantly differentiate NSCLC from SCLC (AUC 0.903 and 0.856, respectively).

CONCLUSIONS: This study is the first to show that a panel of six serum TM can be useful in the clinical management of LC. It may do so either as a first step in the screening of high risk populations (its high sensitivity and NPV can help to exclude LC) and/or through the diagnostic assessment of patients with a clinical suspicion of LC (high specificity, high PPV), even in early stages of the disease. In addition, the six serum TM show a significant capacity to differentiate NSCLC and SCLC, which may require different therapeutic strategies.

PL-39. Role of biomarkers in prostate cancer detection and management

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Cancer of the prostate (PCa) is now recognised as one of the most important medical problems facing the male population. In Europe, PCa is the most common solid neoplasm, with an incidence rate of 214 cases per 1000 men and it is currently the second most common cause of cancer death. Prostate specific antigen (PSA) is a serum marker widely used for the early detection of PCa. Its introduction into clinical practice in the early 1990s had an extraordinary impact on the diagnosis and management of PCa. The PSA-based opportunistic or systematic screening of PCa has resulted in a stage migration to more organ-confined, sometimes indolent tumours at the time of diagnosis. Overdiagnosis and overtreatment remains the main issue in the current PSA driven diagnosis of PCa. Indeed, PSA is not a perfect marker for the detection of PCa because of its low specificity and sensitivity. Considerable efforts have been made to find new PSA markers to accurately detect PCa but also distinguish between clinically significant and insignificant disease and guide the clinical decision making. To this aim, the introduction of several PSA derivatives (free PSA, percentage of free PSA, PSA density, PSA velocity, proPSA) has improved the accuracy of detecting PCa in clinical practice. [-2]proPSA, and Beckman Coulter's Prostate Health Index (PHI), a mathematical algorithm incorporating p2PSA, tPSA and fPSA, defined as: $(p2PSA/fPSA) \times \sqrt{tPSA}$, showed high accuracy for the early detection of PCa.

A multicentre study found, in a cohort study of 646 patients from five European urologic centers with tPSA 2–10 ng/ml who were subjected to initial prostate biopsy for suspected PCa, that ProPSA (p2PSA), %p2PSA, and PHI significantly increased the accuracy of PCa detection avoiding unnecessary biopsies without missing significant cancer.

Preliminary results suggested that the PHI algorithm might also be useful in the follow-up of patients who underwent active surveillance or after radical prostatectomy.

PL-40. Clinical performance of new biomarkers for prostate cancer detection

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Prostate cancer (PCa) detection suffers from low specificity when using PSA alone. The prostate health index (Phi) calculated as $p2PSA/fPSA \times \sqrt{tPSA}$ was evaluated in three different studies.

The first study evaluated Phi in 1362 patients from four different centers with tPSA values 1.6–8.0 µg/L (WHO calibrated). Phi had the largest area under the ROC curve (AUC = 0.74) and performed significantly better as compared with %p2PSA (AUC = 0.72), %fPSA (AUC = 0.61) or tPSA (AUC = 0.56). Significantly higher Phi median values were observed for Gleason score ≥ 7 (Phi = 60) as compared with Gleason score < 7 patients (Phi = 53; $P = 0.002$). The results of this multicenter study demonstrated a superior clinical performance for Phi in detecting PCa, when compared with tPSA or %fPSA. Phi was able to detect aggressive PCa.

The second study compared urinary prostate cancer antigen 3 (PCA3) and TMPRSS2:ERG gene fusion (T2:ERG) with serum Phi in 246 outpatients with PSA values of 0.5–20 µg/L. Of all men 110 (45%) had PCa and 136 men were without PCa. PCA3, Phi and T2: ERG showed the largest AUCs with 0.74, 0.68 and 0.63. PCA3 and Phi were not statistically different. Phi distinguished between Gleason score ≥ 7 (Phi = 62.4) and Gleason score < 7 (Phi = 45.7; $P = 0.001$) patients while both urinary markers failed. Combination of Phi and PCA3 enhanced diagnostic power only modestly.

The third study investigated total testosterone (tT) and its free (fT) and bioavailable (bioT) subforms in serum of 193 men to improve Phi. Phi showed again the largest AUC (0.73) that was additionally increased by the inclusion of bioT or tT in a logistic regression model. The AUC of Phi in patients with tT concentrations of < 8 nmol/L (indicating biochemical hypogonadism) was significantly larger than in patients with higher tT values (0.86 vs. 0.70) showing that the Phi performance could be improved by the simultaneous determination of testosterone.

PL-41. Evaluation of the p2PSA and prostate health index to improve prostate cancer detection

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%p2PSA and Prostate Health Index (phi) have shown valuable results in the detection of prostate cancer (PCa), improving the prediction of the aggressiveness of the tumor. We evaluated %p2PSA and phi in the detection of PCa, estimating their relationship with the aggressiveness of PCa.

354 patients with positive or negative prostate biopsy were included. 150 were enrolled prospectively, and 204 were enrolled retrospectively from our serum bank. Samples were stored at -80°C until the measurements of the tests. PSA, fPSA, and p2PSA serum levels were measured on Access2. %fPSA, %p2PSA and phi were calculated as mathematical combination of these tests. Measurements were estimated and analyzed using SPSS v.18.

The best performance was observed for %p2PSA and phi, obtaining an AUC of 0.723 and 0.732, respectively. The highest specificity at sensitivity around 90% was obtained for phi (27.4%) and %p2PSA (16.2%). Using the cut-off of 31.94 for phi, a reduction of 19% biopsies could be obtained, while 17 PCa would have been missed, including only 4 patients with Gleason score ≥ 7 . Similarly, using a cut-off of 1.21 for %p2PSA, a reduction of 12.7% biopsies could be obtained, while 16 PCa would have been missed, including only 4 patients with Gleason score ≥ 7 . Moreover, among patients with PCa, phi (median: 69.75 vs. 48.04) and %p2PSA (median: 2.60 vs. 1.98) values are significantly higher ($p < 0.0001$) in patients with biopsy Gleason score ≥ 7 .

Our results confirm previous evaluations, showing similar AUCs and results in sensitivity and specificity to other studies. %p2PSA and phi raise the accuracy in the detection of prostate cancer, reducing the number of unnecessary biopsies and improving the prediction of the aggressiveness of the tumor.

PL-42. Benefit of individual risk assessment in early detection of PCa

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Population based screening for prostate cancer detection using PSA is not recommended due to high number of unnecessary biopsies, over-diagnosis and subsequent overtreatment of indolent disease. These drawbacks are related to the lack of specificity and sensitivity of the PSA test to detect clinically significant prostate cancer (PCa). In recent years, several publications demonstrated the benefit of individual multivariate risk assessment using risk calculators in order to reduce unnecessary biopsy and improve detection of clinically significant PCa. On the basis of ERSPC Rotterdam data, in partnership with the prostate cancer research foundation (SWOP) the Rotterdam prostate cancer risk calculator (RPCRC, www.prostatecancer-riskcalculator.com, also available as mobile app) was developed. With this multivariate tool the individual patient risk of detection of prostate cancer can be assessed. A study was conducted to determine the incremental value of the new p2PSA biomarker in the prostate health index (*phi*) incorporated in the RPCRC DRE based risk calculator. A validation set was defined using 615 men with 353 (57%) having PCa with or without previous biopsy from 4 European hospitals. All men had tPSA between 1.6 – 8.0 ng/mL and data on tPSA, freePSA, phi, DRE and prostate volume were used to develop extended versions of the RPCRC RC including the phi index for prediction of PCa at initial biopsy (RC#3_phi) or at repeated biopsy (RC#4_phi) using the likelihood ratio method. The extended RCs were validated on an independent cohort of patients from Milan and Hamburg consisting of 950 men with 404 PCa cases detected. The newly developed RC#3_phi and RC#4_phi showed good discrimination in the validation dataset with significant increase in area under the ROC curves for the detection of clinically significant PCa (clinical stage $> \text{T2b}$ and/or biopsy Gleason score ≥ 7) for men at initial biopsy. The *phi* index improves the prediction ability of the initial risk calculator. Individualised PCa risk assessment using new biomarker and validated risk calculator may help reducing unnecessary biopsy and improves detection of significant PCa, two major limitations of PCa screening.

PL-43. Diagnostic approach to multiple myeloma

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Monoclonal gammopathies are characterized by the proliferation of a single clone of plasma cells that produce a monoclonal protein that can be found in serum, urine or both. The main monoclonal gammopathies include: monoclonal gammopathy of undetermined significance (MGUS), multiple myeloma (MM) and primary systemic amyloidosis (AL amyloidosis). This presentation will focus on the diagnostic approach of MGUS and MM.

Multiple myeloma is a malignant plasma cell proliferation which can result in extensive skeletal involvement with osteolytic lesions or severe osteoporosis, hypercalcemia, anemia, recurrent bacterial infections, and/or extramedullary soft tissue plasmacytomas. In addition, the production of nephrotoxic light chains results in renal function impairment in about 20% of the patients. The annual incidence of MM is 4-5 new cases/100.000 habitants, the median age is 60 years and it is still an

incurable disease with median survivals between 4 and 8 years from diagnosis, despite an improvement in the current therapeutic approaches.

MGUS has a high prevalence (3% and 6% in individuals over 50 and 70 years of age, respectively) and is characterized by the presence of a serum M-protein (< 30 g/L) and less than 10% bone marrow plasma cells with, in contrast to MM, no organ or tissue impairment due to the MG. However, the transformation rate of MGUS to MM is about 1% per year with an actuarial probability of 30% at 25 years of follow-up. The complete work-up for a patient with MM must include all the following: a complete blood cell counts (Hb, platelet and leukocyte counts), complete chemistry study including serum creatinine and calcium, albumin, LDH and beta-2microglobulin, serum and protein studies including total protein measurements plus serum and urine electrophoresis, immunofixation in serum and urine and immunoglobulin measurements. In many instances, free-light chain measurement can be useful for diagnosis, prognosis and/or follow-up. A bone marrow aspirate with an accurate plasma cell count and cytogenetic studies by FISH (del 17p, t(4;14), 1q gains, t(11;14) is mandatory. Finally, a skeletal survey must be performed in all cases while other imaging techniques, such as MRI, CT-scan or PET must be done only when clinically indicated.

PL-44. Diagnostic tools and therapy for primary systemic amyloidosis

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Primary light chain amyloidosis is the most common form of systemic amyloidosis and is caused by misfolded light chains that cause proteotoxicity and rapid decline of vital organ function. Early diagnosis is essential in order to deliver effective therapy and prevent irreversible organ damage. The multiorgan involvement makes these patients particularly fragile and risk assessment is necessary for prognostication and therapeutic decisions. Therapy of AL amyloidosis is highly individualized, risk-adapted and response-tailored. Biomarkers have revolutionized the diagnosis, the risk stratification and therapy of AL amyloidosis.

Diagnosis: The possibility to quantify the amyloidogenic free light chains (FLC) provides a reliable and indispensable clonal marker for the diagnosis of AL amyloidosis. It is also possible to detect early organ involvement using biomarkers: the heart, using the circulating levels of the N-terminal pro-brain natriuretic peptide (NT-proBNP) or BNP, and the kidney, through the quantification of urinary albumin. Periodic measurement of cardiac biomarkers and screening for albuminuria in a patient with MGUS may allow early diagnosis, as abnormalities of either may herald development of amyloidosis, and recovery of organ dysfunction.

Prognosis: Risk-assessment of patients with AL amyloidosis is best achieved using three powerful biomarkers, one indicator of cardiac dysfunction (NT-proBNP), one marker of cardiac damage (cardiac troponin), and the load of cardiotoxic light chain (FLC).

Therapy: The choice of therapy is guided by cardiac biomarkers, NT-proBNP and cardiac troponin. New criteria for response to therapy have been recently defined and validated through an international initiative and are based on modification of FLC and NT-proBNP. The biomarkers-based response criteria, highly correlated with survival outcomes, have been proposed as primary endpoints for clinical trials, accelerating the development of new drugs.

PL-45. Frontline therapy in patients with symptomatic multiple myeloma

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In elderly patients treatment of multiple myeloma (MM) consists of Oral combinations of melphalan and prednisone (MP) plus novel agents are considered as standards of care. The 2 following options are recommended based on data from randomised phase 3 trials: melphalan/prednisone/thalidomide (MPT), or bortezomib/melphalan/prednisone (VMP); both regimens are approved by the European Medicines Agency (EMA).

Lenalidomide combined with low-dose dexamethasone has been compared with MPT in a large randomised phase 3 trial, and has shown promising results, superior to those achieved with MPT in terms of progression-free and OS. Nevertheless, this regimen is not yet approved in Europe.

In younger patients (< 65 years or fit patients in good clinical condition) induction followed by high-dose therapy with autologous stem cell transplantation (ASCT) is the standard treatment. Response rates to induction therapy have been significantly increased by the use of novel agent-based combinations. Bortezomib-dexamethasone has become the backbone of induction therapy prior to ASCT. The addition of a third agent to bortezomib-dexamethasone, e.g. thalidomide (VTD), doxorubicin (DVD or PAD), lenalidomide (RVD) or cyclophosphamide (VCD), has shown higher response rates in phase 2 trials. Three prospective studies have already shown that VTD is superior to TD or bortezomib-dexamethasone. 3-drug combinations including at least bortezomib and dexamethasone are currently the standard of care prior to ASCT.

Tandem ASCT have been evaluated before the era of novel agents. The benefit of tandem ASCT was observed in patients that were not reaching very good partial response (VGPR) after the first ASCT. In recent study from the Netherlands and Germany (Hovon 65-GMMG HD4 trial), in the context of bortezomib induction and maintenance treatment, OS was better in the GMMG group, which performed tandem ASCT in contrast to HOVON (single ASCT). Ongoing trials comparing prospectively single versus tandem ASCT in the era of novel agents will solve this important issue.

Consolidation: Thus far, there is still not enough evidence that consolidation therapy should be systematically applied.

Maintenance: In elderly patients following induction, randomised trials have explored the benefit of maintenance therapy using either IMiDs or bortezomib: MP vs MPR vs MPR-R, VMPT-VT vs VMP, VMP vs VTP followed by either VP or VP maintenance. Due to the trial design, the benefit in OS is not well-established. These drugs are not approved by EMA. Therefore, a systematic maintenance therapy is also not recommended in elderly patients.

In young patients following ASCT, phase 3 randomised trials have demonstrated that maintenance therapy with IMiDs prolongs PFS, but the OS benefit is still unclear. Bortezomib maintenance is also under evaluation. These agents are not approved in this setting; therefore, a systematic maintenance therapy is not recommended.

PL-46. Response assessment and monitoring of patients with multiple myeloma

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Monitoring multiple myeloma (MM) is essential during the evaluation of response to each line of therapy, after transplantation, and at the time of

relapse or progression in all patients. An initial complete workup, including appropriate protein studies in serum and urine, is mandatory. Monitoring of serum and urine total proteins is crucial to establish in a quantitative manner the M-spike, as is electrophoresis immunofixation (IFE) to determine the specific light and heavy chain constituting the monoclonal component.

The use of uniform criteria is particularly important in the context of clinical trials. Complete remission (CR) definition, the goal for the majority of patients, is now in constant evolution. Initially, it was defined as the disappearance of the M-protein by serum and urine IFE, along with the disappearance of plasmacytomas and normal numbers of bone marrow plasma cells. The importance of achieving a CR, defined by negative serum and urine IFE, has been shown in multiple studies.

Nowadays, immunophenotypic and molecular minimal residual disease (MRD) measurement in bone marrow as well as imaging techniques are being standardized. Other sources for MRD determination, i.e. peripheral blood or bone marrow aspiration guided by imaging techniques could be used, reflecting a more accurate and systemic evaluation of residual malignant plasma cells. The other response categories were partial response (PR), that requires a serum M protein decrease of 50% or more and a urine M protein decrease $\geq 90\%$ or more, and/or to < 200 mg/24 hours as well as a reduction in 50% or more in the cross-sectional areas of extraosseous plasmacytomas; and minimal response (MR), requiring a serum M protein decrease of 25% or more and a urine M protein decrease $\geq 50\%$ or more, as well as a reduction in 25% or more in the cross-sectional areas of extraosseous plasmacytomas.

After treatment, identification of relapse/progression with traditional and novel techniques for eventual prompt intervention with rescue treatment is a current issue of debate. Progression and relapse were defined according to the previous response achieved in the patients (less than CR or CR, respectively). Briefly, in patients with PR, an increase of $> 25\%$ with an absolute increase ≥ 5 g/L of the lowest M-component level previously achieved was required for progression. In patients with CR, relapse was defined by recurrence of detectable M-component on IFE, even if electrophoresis remained negative, excluding oligoclonal immune reconstitution. In patients in a treatment-free phase, an every 2-3 months clinical and laboratory follow up is standard.

PL-47. Standardisation of FIT: A Necessary Prerequisite for CRC Screening

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Faecal immunochemical tests for haemoglobin (FIT) are the best non-invasive investigations currently available for colorectal cancer screening. FIT use antibodies to detect intact haemoglobin (Hb) and early degradation products. They have many advantages: usually a single faeces is sampled, specimen collection devices are user-friendly, no dietary or drug restriction is required, and only bleeding in the lower gut is detected. FIT are of two type, qualitative and quantitative, the latter being more useful, since the cut-off faecal Hb concentration used for referral for further investigation can be chosen by programme organisers. However, available FIT are not the same and do not give consistent clinical outcomes. Also, there are major issues with both types of FIT which seriously hinder evaluation of published data and limit transfer of data over time and geography. An Expert Working Group (www.worldendo.org/weo-crcsc-expert-working-group-fit-for-screening.html) has addressed some of the difficulties. Nomenclature must be standardised. The units generally used for both types of FIT are ng Hb/ml buffer and data in these units cannot be transferred between analytical systems: an advocated solution is the

universal adoption of $\mu\text{g Hb/g faeces}$, but this introduces real difficulties in estimation of the mass of faeces collected, faeces having a wide range of consistencies, and the volume of buffer used. Reporting of data in publications on FIT requires harmonisation through use of the new FITTER standard and checklist by investigators and reviewers. Improvement in analytical and clinical evaluations and comparisons of FIT is warranted by adoption of FIT-specific protocols. The specificity of methods to the measurand should be better defined by examination of Hb variants and degradation products. Improvements in calibration of FIT, as per the hierarchy of methods and materials used in laboratory medicine, would also facilitate better comparison of data on FIT.

PL-48. Clinical Utility of HM JACKarc for the Detection of Colorectal Cancer and High-Risk Adenomas

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BACKGROUND: Screening for colorectal cancer can identify the disease at an early stage or at a premalignant phase. Fecal occult blood tests have been developed to detect imperceptible bleeding from colorectal neoplasia. We performed a laboratory and clinical evaluation of HM JACKarc (Kyowa Medex Co., Ltd), an immunoturbidimetric analyzer for the detection of fecal occult blood.

MATERIALS AND METHODS: The imprecision, linearity of dilution and carry-over were determined. We evaluated the efficacy of HM JACKarc to detect advanced colorectal neoplasia (ACRN) by using tissue samples from patients with gastrointestinal symptoms undergoing colonoscopy.

RESULTS (partial data): Imprecision ranged between 3.49% and 6.65%. Linearity upon dilution revealed a mean recovery of all dilutions of 97.33%. No carry-over up to 10,000 ng/mL was detected. Fecal hemoglobin levels were significantly higher in subjects with ACRN (colorectal cancer or high-risk adenoma) (55.7 IQR:1.6–493 ng/mL) when compared to subjects with low-risk adenoma (0.60 IQR:0.30–0.67 ng/mL) ($p < 0.001$), other lesions (1.00 IQR:0.30–2.85 ng/mL) ($p = 0.005$), and normal colonoscopy examination (0.6 IQR:0.25–1.20 ng/mL) ($p < 0.001$). The highest levels were found in patients with colorectal cancer. The AUC of the ROC curve for ACRN was 0.85. The positive predictive value at 30 ng/ml cut-off (30 mg Hb/Kg feces) was 36.4%, the sensitivity 50% and the specificity 94%.

CONCLUSION: HM JACKarc analyzer is easy to handle and safe for personnel to use. Its analytical and clinical performance makes it suitable for use in a clinical chemistry laboratory for the early detection of colorectal cancer and high-risk adenomas.

PL-49. Colorectal cancer screening in Europe and in the Czech Republic

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Colorectal cancer (CRCA) is the second most frequent malignant disease in Europe. CRCA screening began in the Czech Republic in 1994, and the National Colorectal Cancer Screening Program in the Czech Republic started in the year 2000, as a two-step screening program - guaiac based faecal occult blood tests (gFOBT) and colonoscopy, or one-step program with primary screening colonoscopy. Based on the screening colonoscopy database established in 2006, a precise program of evaluation can be

performed using internationally respected quality indicators. The compliance of the Czech populating is growing, having recently reached almost 25% (there is about 10% of screening colonoscopies). Since 2009 also immunochemical FOBT tests (FIT) were also used and the external quality assessment (EQA) of haemoglobin determination in the stool started in January 2012 as a part of the national EQA. This EQA programme is provided by SEKK (member of EQALM) which is accredited according to ISO/IEC 17043:2010 and provides EQA programmes for Czech and Slovak republics. The new FOB EQA programme was joined by 33 clinical laboratories analysing quantitative FIT samples. Since 2013 we have used only immunochemical tests (FIT), and in January 2014 we started population-based screening CRCA using personalized invitation to target population. Quantitative immunochemical detection of occult blood in the stool becomes a primary screening test for colorectal cancer in most European countries, and this review compares the experience of the CRCA screening in the Czech Republic with European experiences. We analysed 152 scientific publications published during the last 5 years (2008–2013) focused on the immunochemical detection of occult blood in the stool. Supported by a Research Project of the General University Hospital Prague RVO-VFN 64165

PL-50. Serum HER-2/neu: Biology and Clinical Utility in Breast Cancer

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Biology: The HER2/neu (erbB2) oncogene encodes a growth factor receptor (p185HER2) that is amplified or overexpressed in 15–20% of breast cancers. The extracellular domain of HER2 can be generated by sheddases (ADAM 10) or alternate transcripts. The remaining truncated HER2 (p95HER2) imbedded in the tumor cell membrane is constitutively active. The Oncogene Science HER2 ELISA was cleared by the FDA in 2000 for monitoring of metastatic breast cancer (MBC) and is elevated (>15 ng/ml) in 9 to 23% of primary breast cancer, and in 22 to 73% of MBC patients.

Elevated serum HER2 predicts reduced response to hormonal therapy: We and others have reported that elevated pretreatment serum HER2 is predictive of decreased response to hormone therapy (tamoxifen and/or aromatase inhibitors). In our study, the 164 of 562 patients with elevated sHER2 (29%) had significantly reduced ORR, CBR, TTP (median 5.7 vs 9.4 mo) and OS (21 vs 37 mo) to first-line hormonal therapy in MBC. Serum HER2 and prediction of response to HER2-targeted therapy in MBC: In tissue HER2-positive MBC, we and others have reported that there is generally no correlation between pretreatment sHER2 and trastuzumab response. But there is some evidence that patients with the highest levels of sHER2 may be less responsive to trastuzumab. However, for lapatinib-treated patients, two phase 3 trials have reported that pretreatment sHER2 did not predict response, whereas a third phase 3 trial of HR-positive MBC reported that patients with elevated pretreatment sHER2 predicted for significantly increased response to lapatinib (HR 0.44 for PFS), but no difference in patients with normal pretreatment HER2 levels. Further study in larger randomized trials is needed to evaluate this differential sHER2 treatment predictive utility. Pretreatment sHER2 has also been reported to be one of the few significant prognostic factors in the CLEOPATRA trial of trastuzumab + pertuzumab in MBC. When evaluating the change in sHER2 levels after treatment, however, a more unified result may be apparent: MBC patients who do not achieve a sHER2 decline of >20% at 2–12 weeks have significantly reduced TTP and OS to trastuzumab-containing therapy;

similarly lapatinib-treated patients not achieving a sHER2 decline of >20% post-treatment had significantly reduced ORR and PFS.

Serum HER2 and prediction of response to HER2-targeted therapy in early breast cancer: In the largest adjuvant trial of trastuzumab and sHER2 reported (NCCTG N9831), an elevated baseline sHER2 level was reported as a prognostic marker associated with shorter DFS, and an elevated sHER2 level at recurrence was predictive of shorter OS (51% compared with 77%) at 3 years.

Conclusion: sHER2 may be useful for predicting response in breast cancer, and for selecting the most effective treatments in the rapidly-expanding arsenal of HER2-targeted therapies.

PL-51. Serum HER2 in prostate cancer

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The HER2 oncogene is involved in the biology of many different tumor types and serves as a prognostic marker and a therapeutic target in breast and gastric cancer. In prostate cancer (PC) studies on HER2 overexpression and gene amplification have yielded different results. Moreover, no benefit of anti-HER2 therapies have yet shown a significant clinical activity in this disease. Despite of that, HER2 seems to play a major role in the oncogenesis of PC, especially in the castration resistant progression. On the other hand, the overexpression of HER2 has been reported to release soluble extracellular domain (ECD) in the serum of PCa patients. In localized PC patients, serum HER2 levels have been correlated with high risk of relapse. Moreover, elevated levels are more frequently observed in metastatic PC. In our experience, high HER2 levels in metastatic castration resistant PC correlated with HER2 tumor staining, lower response to docetaxel-based chemotherapy and impaired overall survival.

PL-52. Circulating Levels of HER-2/neu Oncoprotein in Breast Cancer: Clinical utility as tumor marker and in therapy monitoring

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HER-2/neu, also known as c-erbB-2/neu, is an oncogene located in chromosome 17 that encodes HER-2/neu, a transmembrane protein belonging to the EGFR family. The external domain of this protein is released by the cell and can be studied in serum by immunoassay. HER-2/neu in serum is a specific tumor marker and only slight elevations (> 15 ng/ml) may be found in less than 6% of patients with benign diseases (0.6% > 25 ng/ml), mainly in association with liver diseases. Likewise, the highest concentrations of this oncoprotein are found in patients with advanced breast cancer (sensitivity 42.1% (254/603), median 17, 3 ng/ml), but lower concentrations may be found in other metastatic malignancies (sensitivity: 15% (50/303), median 9.8 ng/ml), particularly ovarian, prostate and lung cancers (mainly adenocarcinomas). HER-2/neu assay sensitivity in patients with untreated primary loco-regional breast cancer is around 11% (163/1523 patients) and seems to be related to overexpression in tissue as well as to the most important prognostic factors: tumor size and nodal involvement. Serial HER-2/neu determinations after surgery seem to be useful in the early diagnosis of recurrence, mainly in patients with HER-2/neu over-expression in tissue, but additional studies are necessary to confirm these results. HER-2/neu sensitivity in patients

with metastasis is around 42.1% (254/603), with a clear relationship to tissue over-expression and to site (higher in visceral metastases) and number of metastases. The clinical utility of HER-2/neu in therapy monitoring of patients with advanced disease is controversial. The best results are obtained using HER-2/neu in tissue to decide Trastuzumab therapy and HER-2/neu and other tumor markers in serum to evaluate the tumor response in primary or secondary chemotherapy regimens.

PL-53. The Italian experience in CRC screening

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The National Screening Monitoring Centre (ONS) is conducting every year on behalf of the Ministry of Health. Here we present a survey of the Italian CRC screening activity, designed to collect information on a standard set of quality indicators, defined together with the Italian group for colorectal cancer screening (GISCoR).

Data for the year 2011 showed that 65% of people aged 50–69 years were living in areas covered by the 105 ongoing organised screening programmes. The proportion of target population covered by the invitation showed a wide variability across the country, ranging from 90% in Northern Italy to 24% in the South.

The majority of the programmes implemented screening with biennial immunochemical faecal occult blood test (FIT), while some adopted sigmoidoscopy (FS) once in a lifetime and FIT for non-responders to FS. Overall, about 3,404,000 subjects were invited to undergo FIT and about 1,568,800 were screened: the adjusted attendance rate was 48%. Large differences in the attendance rate were observed both within and among regions: 10% of programmes reported values lower than 24%.

The positivity rate of FIT programmes was 5.5% at first screening (range: 1.6–11.3%) and 4.3% at repeat screening (range: 3.2–6.7%). The average attendance rate to colonoscopy (TC) assessment was 81.4% and the completion rate of the examinations was 88.7%. Among the 740,281 subjects attending screening for the first time, the detection rate (DR) per 1,000 screened subjects was 2.4 for invasive cancer and 10.3 for advanced adenomas (AA - adenoma $\geq$ 10 mm, or with villous component $>$ 20%, or with high-grade dysplasia). As expected, the corresponding figures in the 843,204 subjects at repeat screening were lower (1.2% and 7.6% for invasive cancer and AA, respectively).

Nine programmes adopted FS as primary screening test: 98% of the target population (about 60,000 subjects) were invited, and 13,629 subjects were screened, with an overall attendance rate of 24%. Considering also the response rate to the offer of FIT to those who did not responded to the FS invitation, the proportion of people undergoing screening in the target age reaches 34%.

Overall, 87% of the FS examinations were classified as complete. TC referral rate was 9.5% and the DR per 1,000 screened subjects was 2.8 and 40.6 for invasive cancer and AA, respectively.

PL-54. FIT and colonoscopy. Competition and cooperation

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The Faecal haemoglobin Immunochemical Test (FIT) represents an improvement in the screening of colorectal cancer (CRC); mainly for its

specificity to quantify specific human haemoglobin and because FIT is feasible for the user and easy to analyze for the laboratory, since the possibility of automation represents an important reduction of the analytical time, and consequently increasing the laboratory capabilities. The Barcelona's CRC screening programme using FIT started in December 2009, is addressed to 197,839 asymptomatic men and women aged 50 to 69 years. The participation rate was 43.5% from the eligible population with a positive test rate (≥ 100 ng/mL or ≥ 20 mg/Kg) of 6.2%. During the first round 1,639 high-risk adenomas (HRA) and 245 invasive CRC were detected. In our opinion the capabilities of the FIT can compete with the colonoscopy. The diagnostic yield of the FIT for the detection of CRC is comparable with colonoscopy, however for the detection of advanced adenoma, colonoscopy is 2.3 fold more effective when compared with the FIT, however this advantage is widely reduced when the compliance and uptake are analysed because the FIT is twice preferred than colonoscopy, therefore in a CRC screening programme the diagnostic yield of the FIT followed by colonoscopy may be higher than the use of colonoscopy alone. Another feature of the FIT is the ability to stratify the individuals in different risk groups in relation to faecal haemoglobin. For example, the risk to detect HRA or CRC in men, 60–69 years old with a result above 886 ng/mL (177 mg/Kg) is 11.5 fold higher when compared with women, 50–59 years old and with a FIT result between 100–158 ng/mL (20 – 32 mg/Kg). Summarizing, The introduction of FIT is an important advance in CRC screening programmes, recent studies show a diagnostic yield comparable with colonoscopy and allow to stratify the individuals with a positive test in different risk groups to detect CRC o HRA.

PL-55. Biomarkers in colorectal cancer

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Colorectal cancer is one of the leading cancer type in the world. Colorectal cancer includes three different genotypes namely chromosomal instability (70–85%), mismatch repair deficiency, with microsatellite instability (10–20%) and hypermethylator phenotype (3–10%).

Beside the already known markers, recently whole genomic experiments are continuously broadening the list of genetic markers. These biomarkers are either to be analysed from stool or from peripheral blood overcoming the patient's acceptance, cost issues, risks against and of the gold standard, colonoscopy.

Chromosomal instability can be identified by DNA aneuploidy and DNA mutation of APC, β -catenin, KRAS, BRAF, TP53 and SMAD4 mutations in 5–45% of the cancers. Microsatellite instability (MSI) can be detected by decreased protein expression of hMSH2, hMSH6, hMLH1 in 35–49% of the MSI cancers. The microsatellite alterations can be characterised by a panel of 3–5 alterations (BATs). Hypermethylator phenotype can be identified by methylated hMLH1, CDKN2A, MINT1, MINT2, MINT 3 or by the alternative panel of CACNA1G, IGF2, NEUROG1, RUNX3, SOCS1. Genotyping of CRC became important due to the different therapeutical response and prognostic outcome of these three cancer genotypes. Therapy response prediction markers became routine part of the colorectal cancer genetic analysis, as of today EGFR, K-RAS, BRAF, PI3CA, AKT mutation determinations are available leading to a more personalized cancer care.

Blood-based screening techniques can also hold diagnostic potential. In blood, mutation of a multigene panel (containing TP53, APC, KRAS, NRAS, HRAS BRAF, ALK, AKT, PIK3CA) can be detected and quantified with high sensitivity for monitoring relapse and therapy

response prediction (OncoBEAM, Inostics Inc). Beside the genetic alterations, epigenetic factors, such as DNA methylation can also be a useful tool for the early and sensitive detection of benign and malign colorectal lesions. The methylated Septin9 is proved to be a 100% tissue marker for colorectal adenoma and cancer. In plasma the sensitivity of the Epi Procolon 2.0 assay (Epigenomics AG) is 85%, specificity and up to 87% regardless of the tumor stage and localisation. Recently, other markers as Syndecan 2 (SDC2) methylation marker was also detected both in tissue and plasma. Further biomarkers, as miRNA and protein markers were also published for colorectal cancer early detection from blood.

For the non-invasive molecular diagnostic of CRC screening the recently developed stool test (Cologuard, Exact Sciences Inc) including a panel of mutational and methylation markers (β -actin, KRAS, BMP3, NDRG4) can be useful in the diagnosis and management of CRC.

Recently discovered specific and sensitive tissue and plasma biomarkers can improve now early diagnostics, screening as well as personalized cancer therapy.

PL-56. PSA-Pyramid in men with low level PSA: a plea for individualized prostate cancer screening (European randomized study of screening for prostate cancer; AARAU)

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Background: Interestingly, various Prostate-Specific Antigen (PSA) screening strategies are being promoted without individual consideration of baseline PSA values. The aim was to investigate rescreening intervals according to baseline PSA of < 3 ng/ml stratified by any and aggressive prostate cancer (PCa), as well as interval PCa incidence.

Patients and Methods: From September 1998 to August 2003, 4350 men with initial PSA-value of < 3.0 ng/ml were prospectively enrolled at our institution. Information on cancer incidences were retrieved through national cancer registries. Rescreening took place every 4 years. A serum PSA of ≥ 3 ng/ml was applied for prostate biopsy indication. Family history (FH) was obtained by a self administered questionnaire at the time of invitation. We analyzed overall PCa detection, interval PCa incidence, tumor grade and disease-specific survival according to baseline PSA. Univariable and multivariable Cox regression analysis tested the influence of age, PSA at baseline, IPSS and FH on PCa incidence and disease-specific survival in order to adapt rescreening intervals.

Results: Baseline PSA of <1.0 vs. 1-1.9 vs. 2-2.9 ng/ml, was present in 2416 (55.5%, group 1, base of the pyramid) vs. 1371 (31.6%, group 2, center of the pyramid) vs. 563 (12.9%, group 3, apex of the pyramid) men, respectively. Stratified according to baseline PSA group, aggressive PCa (Gleason 7-10) was detected in 25 (1.0%) vs. 80 (5.9%) vs. 34 (6.0%) patients. After a median follow-up of 11.6 years, 2 men (0.08%, group 1), 2 (0.14%, group 2) and 1 (0.2%, group 3) men, respectively, died from PCa. Interval PCa surfaced clinically in 0.95% (group1), 3.6% (group2) and 5.2% (group 3), respectively. In multivariable Cox regression analysis, increasing baseline PSA category was an independent risk factor (all $p < 0.0001$) and associated with a 6- to 7-fold elevated risk of detecting aggressive / significant interval PCa (Gleason score ≥ 7).

Conclusion: According to the very low rate of aggressive PCa and interval PCa in these groups after 8 and 4 years, respectively, rescreening intervals can be safely adapted to the baseline PSA values just like a “pyramide”.

PL-57. Liquid biopsy: quantification of the mutation status of plasma DNA in cancer patients

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The concept of “companion diagnostics” in the treatment of cancer patients with “targeted therapies” implies the pretherapeutic therapy stratification on basis of tissue mutation status such as EGFR mutations for TKI therapies in lung cancer and absence of K-Ras mutations for cetuximab therapy in colorectal cancer. However, therapy response rates and duration are still limited due to primary or secondary resistances. Recently genetic heterogeneity including spatial and temporal variability within a tumor or between primary tumor, lymph node and distant metastases was uncovered explaining the difficulties of one-time molecular examinations in tissue biopsy material. Therefore the need for continuous monitoring of the overall mutation status in the patient body as stratification tool for therapy modification became evident.

Nowadays, sensitive blood-based diagnostics identifying mutations in circulating tumor cells (CTC) and in circulating cell-free plasma DNA (cfDNA) are available. They could overcome the genetic heterogeneity as circulating cfDNA reflects the cancerous DNA changes in the whole body. As this concept of “liquid biopsy” is only minimally invasive it can be used to complement tissue biopsy for patient stratification and for the serial monitoring of successfully treated and newly occurring resistant cell clones at an individual level. BEAMing (beads, emulsion, amplification and magnetics) technology allows the sensitive quantification of specific mutations and has recently shown great potential for therapy monitoring, early recurrence detection and resistance monitoring in colorectal and lung cancer. Here we show the highly sensitive detection of K-Ras mutations in plasma DNA of pancreatic cancer. Most remarkably, some patients with progression during chemotherapy had K-Ras-positivity in plasma DNA but not in tumor tissue underlining the potentially high clinical relevance for therapy prediction and prognosis of this “liquid biopsy” approach.

PL-58. Metanephrines in pheochromocytoma - only biomarkers or predictors of genetic background?

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Pheochromocytomas and paragangliomas are tumors arising from adrenal or extra-adrenal chromaffin tissue. Pheochromocytomas and sympathetic paragangliomas present usually with typical paroxysmal symptoms but may be first diagnosed as incidentally found masses. Biochemical diagnosis represents a crucial step in the diagnostic workup as undiagnosed tumors are associated with cardiovascular morbidity or risk of metastases. Previously used markers such as vanillylmandelic acid or catecholamines have been substituted with metanephrines, which are O-methyl metabolites of catecholamines produced only in the adrenal medulla or tumorous tissue. Metanephrines are secreted continuously compared to maternal catecholamines and so have better sensitivity and specificity. Metanephrine levels correlate with the tumor mass. Increased levels of plasma methoxytyramine may point to the risk of malignancy.

More than 10 genes have been discovered whose mutations are pathogenic for pheochromocytoma or paraganglioma. The type of secreted catecholamines may also predict the result of genetic testing – epinephrine/metanephrine secreting tumors associated with multiple endocrine neoplasia type 2 or neurofibromatosis type I, and normetanephrine/norepinephrine secreting tumors in von Hippel-Lindau syndrome or due to mutations several subunits of the succinate dehydrogenase gene. Elevation of plasma dopamine or methoxytyramine levels is often found in subjects with mutated B or D subunit of succinate dehydrogenase gene and may be the only one elevated biomarker in these patients. The biochemical phenotype also predicts usefulness of particular methods for functional imaging of these tumors – more specific radionuclides such as ^{18}F -fluorodopamine provide better results in tumors with mature secretory phenotype (production of epinephrine) than in tumors with immature phenotype (norepinephrine or only dopamine) in which ^{18}F -fluorodeoxyglucose is the radionuclide of choice.

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PL-59. New Markers for Lymphomas, Leukemias and Myeloma - Thymidine Kinase, sIL-2R, b2Microglobulin and IL-6. Thymidine Kinase can discriminate patients with a Low to High Grade Lymphoma transformation

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Introduction: Hematological malignancies account for about 10% of all cancers which can be divided into slow growing Low grade or rapidly growing High Grade diseases, differing in prognosis and therapies.

Study Aims: To assess the clinical potential of the new TK assay, sIL-2R, IL-6 and b2M in Hematological Neoplasms and correlate levels with baseline parameters, overall survival and ability to discriminate between Low to High Grade Lymphomas.

Patients and methods: We analyzed a total of 198 patients (pts) using DiaSorin TK1 ELISA tested during the treatment initiation at diagnosis or relapse, except for TK kinetics. Most pts had DLBCL (60 including transformation) and Low Grade NHL (46), in addition pts with HL, MM and CLL, as well as with very aggressive tumors such as Burkitt's Ly/Leukemia, ALL and AML. We compared a cohort of pts with transformation of Low to High Grade Lymphoma and pts with Low Grade Lymphoma with suspected transformation with other diagnoses (i.e. infections, low grade related symptoms, large low grade-related masses, etc).

Results: we first evaluated TK, B2M, and LDH in 49 normal adults. The mean value was 4.06 u/L $\pm$ 1.83, but we set the upper normal limit to 7.7 u/L (average + 2SD). Low Grade Lymphomas had significantly ($p < 0.03$) lower readings, 12.8 u/L ($\pm$ 21.5) compared to 32.8 ($\pm$ 36.6) u/L for High Grade diseases. In Low Grade Diseases TK levels did not correlate with survival. However, in High Grade Diseases TK levels correlated with a significantly better long term survival ($p = 0.04$). To notice, 6 pts with transformations had TK levels up to 48.8 u/L ($\pm$ 29.73). We thus further analyzed another cohort of 60 pts with a base line Low Grade Lymphoma and suspected transformation. Of these 36 pts had a suspected condition eventually proving to be Low Grade disease or infection-related and 24 with biopsy proven transformations. The TK differences were very significant ($p < 0.01$) between the 2 groups. However, after censoring for 3 CLL pts with very high WBC count in the Low Grade group and 5 pts with stage 1 in-situ transformations, these differences became striking ($P < 0.0001$) with a median level of 7.6 versus 48.6 u/L, respectively.

A TK level of > 14.1 u/L had a Sensitivity of 95% and a Specificity of 76% for a Low to a High Grade Lymphoma transformation, with a Negative Predictive Value of 96% and a Positive Predictive Value of 69%, respectively. We demonstrated extremely high sIL-2R levels (1,450-18,337 u/ml) in HCL and increasing levels from Low to Intermediate and High Grade Lymphomas. We also found very high IL-6 (18-1,879 pg/ml) levels in MM which could diagnose the disease and predict recurrence. These markers were more sensitive than b2M.

Conclusions: TK levels are useful in assessing prognosis, especially in High Grade Lymphomas, and better than B2M or LDH. Moreover, TK levels were able to discriminate Low Grade pts suspected of a High Grade transformation, but eventually having none and pts with proven transformations. sIL-2R is a useful marker for HCL and IL-6 in MM.